CHAPTER 1

Stochastic convergence

Asymptotic Statistics · MATH 598 · Mehdi Dagdoug ·

In this chapter, we define four important modes of stochastic convergence: almost sure, in probability, in L^p and in distribution. Some of these modes of convergence are stronger than others; we will discuss these relationships. Although some of these modes of convergence are weaker, they sometimes can be combined with other arguments to imply their stronger counterparts; we call these implications partial converses. Often, the convergence properties of various difficult statistics can be deduced by establishing the convergence of several easier statistics, before combining them. Tools like the continuous mapping theorem and Slutsky will be derived. We then conclude by introducing popular (and convenient) stochastic order notations and their algebra.

Notations

Throughout the course, we work on a probability space $(\Omega, \mathcal{A}, \mathbb{P})$. A random vector X is a measurable map $X : \Omega \to \mathbb{R}^d$ for $d \geq 1$; we say that it is a random variable if $d = 1$. To be most useful, our definitions will often be provided for random vectors and we will see random variables as particular cases. When nothing is lost, we will sometimes work with random variables directly. The dimension should be clear from the context.

The random vector X induces a new law[1] $\mathbb{P}_X := \mathbb{P} \circ X^{-1}$ on $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$, that is,

$$\mathbb{P}\left[X \in A\right] = \mathbb{P}[\{\omega \in \Omega : X(\omega) \in A\}] = \mathbb{P}\left[X^{-1}(A)\right] = \mathbb{P}_X(A), \qquad A \in \mathcal{B}(\mathbb{R}^d),$$

where $\mathcal{B}(\mathbb{R}^d)$ denotes the Borel sigma-algebra[2] on $\mathbb{R}^d$.

We will denote by $\|x\| := (\sum_{j=1}^d x_j^2)^{1/2}$, the Euclidean norm of a vector $x \in \mathbb{R}^d$. The distribution function of a random vector X is denoted

$$F_X : t \mapsto \mathbb{P}[X \leq t] = \mathbb{P}[X_{(1)} \leq t_{(1)}, ..., X_{(d)} \leq t_{(d)}], \qquad t \in \mathbb{R}^d.$$

The law of X is denoted by $\mathcal{L}(X)$. For example, if $X \sim \mathcal{N}(0,1)$, then $\mathcal{L}(X) = \mathcal{N}(0,1)$. The expectation of X, defined when $\mathbb{E}[\|X\|] < \infty$, is given by

$$\mathbb{E}[X] = \int_\Omega X(\omega)\mathbb{P}(\mathrm{d}\omega) = \int_{\mathbb{R}^d} x\mathbb{P}_X(\mathrm{d}x).$$

[1] In this text, the word "law" is synonymous with "distribution".

[2] A Borel sigma-algebra on a (topological) space E is the sigma-algebra generated by the open sets of E.

1.1 The main modes of stochastic convergence

The starting point of our discussion on convergence of random variables is to notice that random variables are functions, in the world of probability. We typically will say that a sequence of vectors $(x_n)_{n\in\mathbb{N}}$ converges to a point x if for all $\epsilon > 0$, there exists an integer $n_0 := n_0(\epsilon) \in \mathbb{N}$ such that, for all $n \geq n_0$, $\|x_n - x\| \leq \epsilon$. That is, for any given threshold (ϵ), there exists a moment (n_0) after which, the sequence $(x_n)_{n\in\mathbb{N}}$ will stay close to x (within an ϵ neighborhood).

However, *random variables are functions*. Two types of convergence of sequences of functions are most common: pointwise and uniform convergence. Let $(f_n)_{n\in\mathbb{N}}$ be a sequence of functions $f_n : A \to \mathbb{R}^d$ and $f : A \to \mathbb{R}^d$ be another function. We say that f_n converges pointwise to f on A if, for every fixed $x \in A$, the sequence $f_n(x)$ converges to $f(x)$ (note, this is now a sequence of vectors). Uniform convergence of f_n to f, on the other hand, means that $\sup_{x\in A}\|f_n(x) - f(x)\| \to 0$ as $n \to \infty$. Uniform convergence is stronger than pointwise convergence (it implies it). However, although these notions of convergence of sequences of functions are useful, *random variables are functions in a world of probability*. We will therefore combine convergence of functions with probability statements.

Although more types of convergence of random variables exist, in statistics, only four types will be useful in these notes: almost sure convergence, convergence in probability, in L^p, and in law. These will be the focus of this section.

1.1.1 Almost sure convergence

Almost sure convergence is perhaps the most natural way to define convergence of a sequence of random variables, in that it is a natural extension of the deterministic setting. Almost sure convergence is also called convergence with probability one[3].

Definition 1.1 (Almost sure convergence).
A sequence of random vectors $(X_n)_{n\in\mathbb{N}}$ ***converges almost surely*** to a random vector X, denoted $X_n \xrightarrow[n\to\infty]{a.s.} X$, if

$$\mathbb{P}\left[\omega : \lim_{n\to\infty} X_n(\omega) = X(\omega)\right] = 1.$$

There are various ways to rewrite the definition of almost sure convergence. The next lemma gives useful equivalences. Before proceeding, we introduce the following notions: infinitely often, finitely many, and limits of sets. These will be needed to further describe almost sure convergence.

Exercise 1.1 (★★).
Let $(A_n)_{n\in\mathbb{N}}$ be a sequence of events.

(a) Show that if $(A_n)_{n\in\mathbb{N}}$ is monotone (increasing or decreasing), then $\liminf_{n\to\infty} A_n = \limsup_{n\to\infty} A_n$, which are equal to their previous definitions (the union for increasing sequences, the intersection for decreasing).

[3] In general, "with probability one" (wp1), "almost surely" (a.s.), or even "almost everywhere" (a.e.), all have the same meaning. Specifically, a statement S is said to hold a.s. on $(\Omega, \mathcal{A}, \mathbb{P})$ if there exists an event $E \in \mathcal{A}$ such that S holds for all $\omega \in E$ and $\mathbb{P}(E) = 1$.

(b) Show that $\{A_n \text{ f.o.}\} = \{A_n \text{ i.o.}\}^c$ and that $\{A_n \text{ a.a.}\} = \{A_n^c \text{ f.o.}\}$, that is,

$$\left(\limsup_{n\to\infty} A_n\right)^c = \liminf_{n\to\infty} A_n^c.$$

(c) Let $A_n := (-1/n, 1]$ when n is even and $A_n := (-1, 1/n]$ when n is odd. Compute $\liminf_{n\to\infty} A_n$ and $\limsup_{n\to\infty} A_n$.

Whenever the sequence of events $(A_n)_{n\in\mathbb{N}}$ converges, ***continuity of the probability*** holds, meaning

$$\lim_{n\to\infty} \mathbb{P}[A_n] = \mathbb{P}\left[\lim_{n\to\infty} A_n\right].$$

Lemma 1.2 (Almost sure convergence: equivalences)**.**
The following statements are equivalent.

(i) $X_n \xrightarrow[n\to\infty]{a.s.} X$.

(ii) There exists an event $E \in \mathcal{A}$, such that $\mathbb{P}(E) = 1$ and X_n converges to X pointwise on E.

(iii) For all $\epsilon > 0$, $\mathbb{P}\left[\|X_n - X\| > \epsilon \text{ i.o.}\right] = 0$.

(iv) For all $\epsilon > 0$, $\lim_{n\to\infty} \mathbb{P}\left[\sup_{k\geq n}\|X_k - X\| > \epsilon\right] = 0$.

Proof. Proof of (i) ⇔ (ii). We let $E := \{\omega \in \Omega : \lim_{n\to\infty} X_n(\omega) = X(\omega)\}$. By definition $X_n \xrightarrow[n\to\infty]{a.s.} X$ if and only if $\mathbb{P}(E) = 1$. Moreover, on E, we have $\lim_{n\to\infty} X_n(\omega) = X(\omega)$, which is the definition of pointwise convergence of X_n to X on E.

Proof of (i) ⇔ (iii). Let $\epsilon > 0$ and define $A_\epsilon := \{\|X_n - X\| > \epsilon, \text{ i.o.}\}$. We will use the following fact.

Fact. A sequence $(u_n)_{n\in\mathbb{N}}$ of nonnegative numbers satisfies $\lim_{n\to\infty} u_n = 0$ iff, for every $\epsilon > 0$, $u_n > \epsilon$ only finitely many times.

Leveraging this fact, we see that

$$\{\|X_n - X\| \to 0\} = \bigcap_{\epsilon>0} A_\epsilon^c.$$

Technique. Observe that

$$\bigcap_{\epsilon>0} A_\epsilon^c = \bigcap_{m\geq 1} A_{1/m}^c.$$

Replacing the uncountable intersection over all $\epsilon > 0$ by a countable intersection is interesting because countable intersections are measurable, uncountable need not be. Here, therefore, convergence to 0 can be tested on the countable set $\{1/m : m \in \mathbb{N}^*\}$.

Finally, to conclude, we leverage the following fact.

Fact. If a countable intersection has probability one, each event still has probability one.

Therefore, we have

$$\begin{aligned}\mathbb{P}\left[\|X_n - X\| \to 0\right] = 1 \quad &\Leftrightarrow \quad \mathbb{P}\left[\bigcap_{m\geq 1} A_{1/m}^c\right] = 1 \\ &\Leftrightarrow \quad \mathbb{P}\left[A_{1/m}\right] = 0, \quad \text{for all } m \geq 1, \\ &\Leftrightarrow \quad \mathbb{P}\left[A_\epsilon\right] = 0, \quad \text{for all } \epsilon > 0.\end{aligned}$$

Proof of (iii) ⇔ (iv). Let $\epsilon > 0$ and $B_n := \{\sup_{k\geq n}\|X_k - X\| > \epsilon\}$, and note that

$$B_n = \bigcup_{k\geq n}\{\|X_k - X\| > \epsilon\}.$$

The events B_n are decreasing, so $B_n \downarrow \cap_{n\geq 1} B_n$. Hence, by continuity from above,

$$\lim_{n\to\infty}\mathbb{P}[B_n] = \mathbb{P}\left[\bigcap_{n\geq 1} B_n\right] = \mathbb{P}\left[\bigcap_{n\geq 1}\bigcup_{k\geq n}\{\|X_k - X\| > \epsilon\}\right] = \mathbb{P}\left[\|X_n - X\| > \epsilon \text{ i.o.}\right].$$

■

Intuitively, the idea of almost sure convergence is that, for almost all realized outcomes ω, the numerical sequence $X_1(\omega), X_2(\omega), X_3(\omega), \ldots$ converges to $X(\omega)$; this is therefore a *pathwise* notion of convergence. So, with probability one, X_n is eventually close to X, and stays close forever.

Exercise 1.2 (★★).
Let $U_1, U_2, \ldots$ be independent random variables with law $\mathcal{U}([0,1])$ and let $M_n := \max_{1\leq k\leq n} U_k$.

(a) Show that, for almost every $\omega \in \Omega$, the sequence $(M_n(\omega))_{n\in\mathbb{N}}$ is nondecreasing and bounded, hence converges to some $M_\infty(\omega) \in [0,1]$.

(b) For $\epsilon \in (0,1)$, compute $\mathbb{P}[M_n \leq 1-\epsilon]$.

(c) Show that $M_n \xrightarrow[n\to\infty]{a.s.} 1$. *Hint:* write $\{M_\infty \leq 1-\epsilon\}$ as a decreasing limit of events, then use $\{M_\infty < 1\} = \bigcup_{m\geq 1}\{M_\infty \leq 1 - 1/m\}$.

1.1.2 Convergence in probability

We now turn to another, very important, notion of convergence of random variables.

Definition 1.3 (Convergence in probability).
A sequence of random vectors $(X_n)_{n\in\mathbb{N}}$ ***converges in probability*** to a random vector X, denoted $X_n \xrightarrow[n\to\infty]{\mathbb{P}} X$, if for all $\epsilon > 0$,

$$\lim_{n\to\infty}\mathbb{P}\left[\|X_n - X\| > \epsilon\right] = 0.$$

Intuitively, convergence in probability means that, for large n, X_n is very likely to be close to X. It is not a pathwise statement, but says that the probability of a large error is small, as n gets large.

Some examples. As a simple example, consider the case where $Z \sim \mathcal{N}(0,1)$ and $X_n := X + Z/n$. Then, the claim is that $X_n \xrightarrow[n\to\infty]{\mathbb{P}} X$. To see this, let $\epsilon > 0$. Then,

$$\mathbb{P}\left[|X_n - X| > \epsilon\right] = \mathbb{P}\left[|Z| > n\epsilon\right].$$

Now, as $n \to \infty$, we have $n\epsilon \to \infty$ so that the probability of $|Z|$ being greater than $n\epsilon$ decreases to 0. To see this more formally, note that the sequence of events $(\{|Z| > n\epsilon\})_{n\in\mathbb{N}}$ is decreasing to $\cap_{n\in\mathbb{N}^*}\{|Z| > n\epsilon\} = \varnothing$, and thus, by continuity from above,

$$\lim_{n\to\infty} \mathbb{P}\left[|Z| > n\epsilon\right] = \mathbb{P}\left[\lim_{n\to\infty}\{|Z| > n\epsilon\}\right] = \mathbb{P}\left[\varnothing\right] = 0.$$

This shows therefore that convergence in probability holds. We plot the densities of X_n around X for various choices of n with two different threshold boxes of ϵ in Figures 1.1 and 1.2.

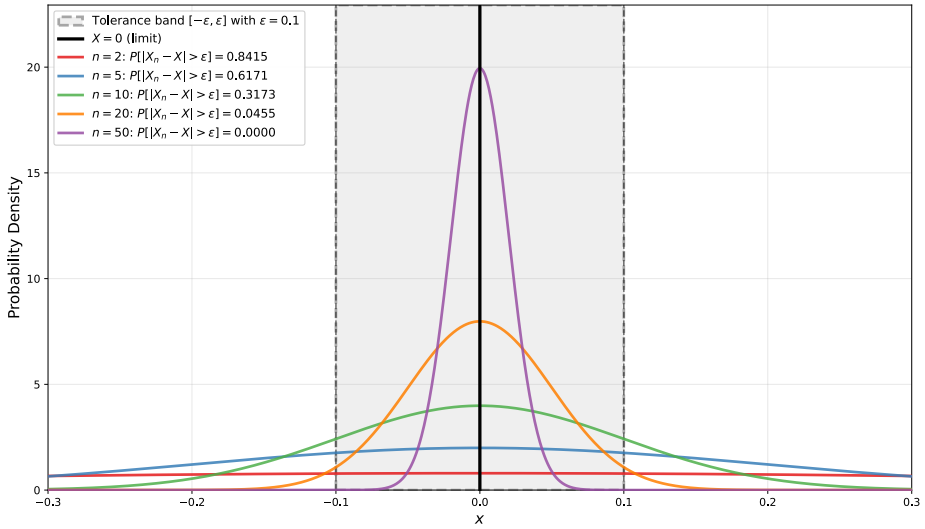


Figure 1.1. Varying densities of X_n around $X = 0$ as n varies with $\epsilon = 0.1$.

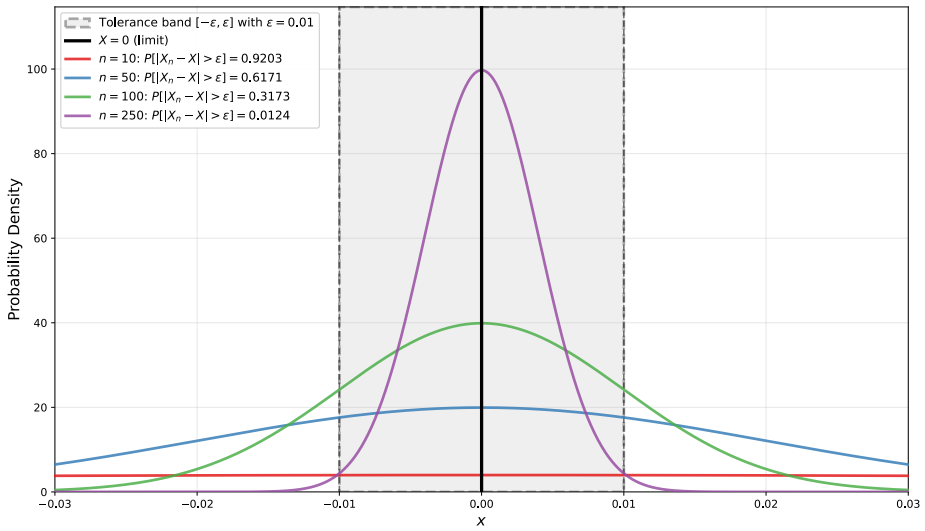


Figure 1.2. Varying densities of X_n around $X = 0$ as n varies with $\epsilon = 0.01$.

It is important to note that it does not forbid rare extreme values. For example, consider the sequence

$$X_n = \begin{cases} n, & \text{with probability } \dfrac{1}{n}, \\ 0, & \text{with probability } 1 - \dfrac{1}{n}. \end{cases}$$

Then, for $\epsilon > 0$ and sufficiently large n, we have

$$\mathbb{P}\left[|X_n - 0| > \epsilon\right] = \mathbb{P}\left[X_n = n\right] = n^{-1},$$

which shows that $X_n \xrightarrow[n\to\infty]{\mathbb{P}} 0$. However, it is also true that, with vanishingly small probability, X_n can take larger and larger values as $n \to \infty$. As we will see later on, this may or may not prevent other types of convergence from happening.

1.1.3 Convergence in distribution

We now introduce a weaker type of convergence, with many important uses in statistics: convergence in distribution. It is also called ***convergence in law*** or ***weak convergence***. To that aim, let us introduce the following notations. Recall that F_X denotes the cumulative distribution function of a random vector X. For a function f, we write $C(f) := \{t \in \mathrm{dom}(f) : f \text{ is continuous at } t\}$ the set of its continuity points. Moreover, we write $C_b(\mathbb{R}^d)$ to denote the set of bounded continuous functions on $\mathbb{R}^d$.

Definition 1.4 (Convergence in distribution).
A sequence of random vectors $(X_n)_{n\in\mathbb{N}}$ ***converges in distribution*** to a random vector X, denoted $X_n \xrightarrow[n\to\infty]{\mathcal{L}} X$, if

$$\lim_{n\to\infty} F_{X_n}(t) = F_X(t), \quad \text{for all } t \in C(F_X),$$

where $C(F_X) := \{t \in \mathrm{dom}(F_X) : F_X \text{ is continuous at } t\}$.

Exercise 1.3 (★★).

(a) Suppose that $X_n \xrightarrow[n\to\infty]{\mathbb{P}} X$ and $X_n \xrightarrow[n\to\infty]{\mathbb{P}} Y$. Show that $\mathbb{P}[X = Y] = 1$.

(b) Suppose that $X_n \xrightarrow[n\to\infty]{a.s.} X$ and $X_n \xrightarrow[n\to\infty]{a.s.} Y$. Show that $\mathbb{P}[X = Y] = 1$.

(c) Let $X \sim \mathcal{N}(0, 1)$ and $X_n := X$ for all n. Show that $X_n \xrightarrow[n\to\infty]{\mathcal{L}} X$ and $X_n \xrightarrow[n\to\infty]{\mathcal{L}} -X$, although $\mathbb{P}[X = -X] = 0$. What is the correct uniqueness statement for convergence in distribution?

Convergence in distribution is somewhat separate from the previous types of convergence, in that it does not require X_n to get close to X in the traditional sense. First, we can note that the definition does not require X_n and X to be defined on the same probability space. Even if they were, it also does not require X_n to be close to X, but that *the distribution of X_n is getting closer to that of X*. It is therefore really a statement about the law of X_n resembling more and more that of X.

Specifically, convergence in distribution requires pointwise convergence of the sequence of functions $(F_{X_n})_{n\in\mathbb{N}}$ to F_X on the set $C(F_X)$. Therefore, pointwise convergence is required only at continuity points of the target CDF F_X.

Some examples. First, let $X_n = -X$ where $X \sim \mathcal{N}(0,1)$. Then, although surprising, $X_n \xrightarrow[n\to\infty]{\mathcal{L}} X$. Indeed, because X is standard Gaussian, its distribution is symmetric[4], so that $\mathcal{L}(X_n) = \mathcal{L}(-X) = \mathcal{L}(X)$, and thus, for any $n \in \mathbb{N}$, for any $t \in \mathbb{R}$, $F_{X_n}(t) = F_X(t)$. However, when $X(\omega) = x$, $X_n(\omega) = -x$, so X_n can be quite far from X, and this for any n.

Another simple example is the following. Let $X_n = X + n^{-1}$, for some random variable X. Then, let $t \in C(F_X)$ and write

$$F_{X_n}(t) = \mathbb{P}\left[X_n \le t\right] = \mathbb{P}\left[X \le t - n^{-1}\right] = F_X(t - n^{-1}).$$

Then, $t - n^{-1} \to t$ as $n \to \infty$ and since t is a continuity point of F_X, we get $\lim_{n\to\infty} F_X(t - n^{-1}) = F_X(t)$, which shows convergence in distribution. We plot the CDFs of X_n and X with varying sample sizes in Figure 1.3.

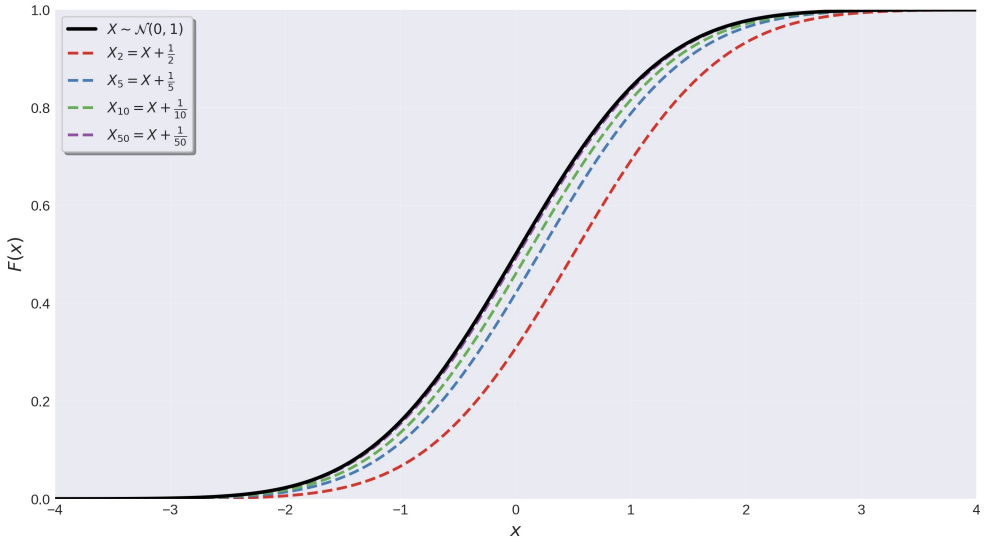


Figure 1.3. Varying CDFs of X_n and X as n varies.

Equivalent reformulations. Although we defined convergence in distribution using CDFs, there are various other possibilities to define it. A well-known lemma, the ***Portmanteau Lemma***, gives several useful equivalences to convergence in distribution.

Before proceeding, let us recall briefly some topological notions. For $x \in \mathbb{R}^d$ and $r > 0$, we denote by $\mathcal{B}(x,r) := \{y \in \mathbb{R}^d : \|x - y\| < r\}$ the *open ball* centered at x with radius r. We say that a set $A \subseteq \mathbb{R}^d$ is *open* if, for every $x \in A$, there exists $r > 0$ such that $\mathcal{B}(x,r) \subset A$. The *interior* of a set A is

$$A^\circ := \{x \in A : \text{there exists } r > 0 \text{ such that } B(x,r) \subseteq A\}.$$

[4] A random variable X has a symmetric distribution (around 0) if $X \overset{d}{=} -X$, that is, the law of X is the same as that of $-X$, i.e., $\mathcal{L}(X) = \mathcal{L}(-X)$. A normal distribution is symmetric around μ.

Thus, A° is the largest open set contained in A. The set A is *closed* if its complement A^c is open. The *closure* of A is the set

$$\overline{A} := \left\{ x \in \mathbb{R}^d : B(x,r) \cap A \neq \varnothing \text{ for every } r > 0 \right\}.$$

Thus, $\overline{A}$ is the smallest closed set containing A. Informally, it is meant to contain the points of A, and those that can be approached arbitrarily closely by points of A. Equivalently, a point $x \in \overline{A}$ if and only if there exists a sequence $(x_n)_{n\in\mathbb{N}} \subset A$ such that $\lim_{n\to\infty} x_n = x$. The *boundary* of A is

$$\partial A = \overline{A} \setminus A^\circ.$$

For example, $(a,b)^\circ = (a,b)$, $\overline{(a,b)} = [a,b]$, and $\partial(a,b) = \{a,b\}$.

Lemma 1.5 (Portmanteau).
The following statements are equivalent.

(i) $X_n \xrightarrow[n\to\infty]{\mathcal{L}} X$.

(ii) For every $f \in C_b(\mathbb{R}^d)$,
$$\lim_{n\to\infty} \mathbb{E}\left[f(X_n)\right] = \mathbb{E}\left[f(X)\right].$$

(iii) For every closed set $F \subseteq \mathbb{R}^d$,
$$\limsup_{n\to\infty} \mathbb{P}\left[X_n \in F\right] \leq \mathbb{P}\left[X \in F\right].$$

(iv) For every open set $G \subseteq \mathbb{R}^d$,
$$\liminf_{n\to\infty} \mathbb{P}\left[X_n \in G\right] \geq \mathbb{P}\left[X \in G\right].$$

(v) For every Borel set $A \subseteq \mathbb{R}^d$ such that $\mathbb{P}\left[X \in \partial A\right] = 0$, we have
$$\lim_{n\to\infty} \mathbb{P}\left[X_n \in A\right] = \mathbb{P}\left[X \in A\right].$$

Proof. See van der Vaart (1998), page 6. ■

Convergence in distribution therefore controls averages of continuous bounded functions. Here, continuity is important because a small change in x only produces a small change in $f(x)$. By contrast, probabilities of events can be written $\mathbb{P}[X_n \in A] = \mathbb{E}[\mathbb{1}_A(X_n)]$, and the indicator $\mathbb{1}_A$ is not continuous: it jumps from 0 to 1 on the boundary of A. By the Portmanteau Lemma, item (v), we see that convergence in distribution does give convergence of probabilities for an event A, provided that the limiting distribution puts no mass on the boundary of A. When the limiting distribution may put probability on the boundary, exact convergence may not hold; the Portmanteau lemma still provides bounds that may be useful.

Exercise 1.4 (★★★).
Let $(X_n)_{n\in\mathbb{N}}$ and X be random variables with values in $\mathbb{N}$.

(a) Show that $X_n \xrightarrow[n\to\infty]{\mathcal{L}} X$ if and only if $\lim_{n\to\infty} \mathbb{P}[X_n = k] = \mathbb{P}[X = k]$ for every $k \in \mathbb{N}$.

(b) Let $\lambda > 0$ and, for $n > \lambda$, let X_n have the Binomial law with parameters n and λ/n. Show that $X_n \xrightarrow[n\to\infty]{\mathcal{L}} X$, where X has the Poisson law with parameter λ, that is, $\mathbb{P}[X = k] = e^{-\lambda}\lambda^k/k!$ for $k \in \mathbb{N}$.

1.1.4 L^p convergence

Before defining convergence in L^p, it is useful to define L^p as a space of random variables. Let $p \geq 1$. The space L^p consists of all random variables whose p-th moment is finite,

$$L^p := \{X : \mathbb{E}\left[\|X\|^p\right] < \infty\}.$$

On the space L^p, we can define a norm $\|\cdot\|_{L^p}$ by

$$\|X\|_{L^p} = \mathbb{E}\left[\|X\|^p\right]^{1/p}.$$

The power $1/p$ is needed for the homogeneity of the norm. We restrict our attention to the case $p \geq 1$ because for $p < 1$, the triangle inequality fails and $\|\cdot\|_{L^p}$ is not a norm. The space L^p is a *Banach space*[5] where random vectors that agree almost surely are identified. The norm itself induces a distance between random vectors,

$$d_{L^p}(X, Y) = \|X - Y\|_{L^p} = \mathbb{E}\left[\|X - Y\|^p\right]^{1/p}.$$

Convergence in L^p is simply convergence with respect to this distance.

Definition 1.6 (Convergence in L^p).
Let $p \geq 1$, $X_n \in L^p$ and $X \in L^p$. A sequence of random vectors $(X_n)_{n\in\mathbb{N}}$ ***converges in L^p*** to a random vector X, denoted $X_n \xrightarrow[n\to\infty]{L^p} X$, if

$$\lim_{n\to\infty} \mathbb{E}\left[\|X_n - X\|^p\right] = 0.$$

Convergence in L^q implies convergence in L^p for all exponents $p < q$. To see this, let us introduce ***Hölder's inequality***: for exponents $p, q \in]1, \infty[$ such that $1/p + 1/q = 1$, random variables $X \in L^p$ and $Y \in L^q$ satisfy

$$\left|\mathbb{E}\left[X^\top Y\right]\right| \leq \mathbb{E}\left[\|X\|^p\right]^{1/p} \mathbb{E}\left[\|Y\|^q\right]^{1/q}.$$

Another very useful inequality is ***Jensen's inequality***: for a random vector $X \in \mathbb{R}^d$ and a convex[6] function $\phi : \mathbb{R}^d \to \mathbb{R}$, we have

$$\phi\left(\mathbb{E}\left[X\right]\right) \leq \mathbb{E}\left[\phi\left(X\right)\right].$$

As a first application of Jensen's inequality, it is worth noting that convergence in L^1 implies convergence of the first moment. Indeed, since $\phi : x \mapsto |x|$ is convex, by Jensen's inequality, we have

$$\left\|\mathbb{E}[X_n] - \mathbb{E}[X]\right\| = \phi\left(\mathbb{E}\left[X_n - X\right]\right) \leq \mathbb{E}\left[\phi\left(X_n - X\right)\right] = \mathbb{E}\left[\left\|X_n - X\right\|\right] \to 0$$

as $n \to \infty$ if $X_n \xrightarrow[n\to\infty]{L^1} X$.

[5] A Banach space is a normed vector space that is complete (i.e., in which every Cauchy sequence converges).

[6] A function f is convex if $f\left(\theta x + (1-\theta)y\right) \leq \theta f(x) + (1-\theta)f(y)$, for all $\theta \in [0, 1]$ and all vectors $x, y \in \mathrm{dom}(f)$.

Theorem 1.7.
Let $p, q \geq 1$ with $p < q$. Then, $L^q \subset L^p$. Moreover, if $X_n \xrightarrow[n\to\infty]{L^q} X$, then $X_n \xrightarrow[n\to\infty]{L^p} X$.

Proof. Let $\phi : x \mapsto \|x\|^{q/p}$. Then, differentiating on $]0, \infty[$ gives

$$\phi^{(2)}(x) = \frac{q}{p}\left(\frac{q}{p} - 1\right) x^{q/p-2} \geq 0, \quad \text{if } q/p \geq 1,$$

which shows that ϕ is convex on $]0, \infty[$ when $q/p \geq 1$. We apply Jensen's inequality on the random variable $\|X\|^p$ to get

$$\left(\mathbb{E}\left[\|X\|^p\right]\right)^{q/p} \leq \mathbb{E}\left[\left(\|X\|^p\right)^{q/p}\right] = \mathbb{E}\left[\|X\|^q\right],$$

and applying the increasing function $x \to x^{1/q}$ on both sides of the above inequality gives

$$\|X\|_{L^p} \leq \|X\|_{L^q}.$$

From this, we see that (i) if $\|X\|_{L^q} < \infty$, then $\|X\|_{L^p} < \infty$, which shows that if $X \in L^q$ then $X \in L^p$; (ii) if $\lim_{n\to\infty}\|X_n - X\|_{L^q} = 0$, then $\lim_{n\to\infty}\|X_n - X\|_{L^p} = 0$ since

$$0 \leq \|X_n - X\|_{L^p} \leq \|X_n - X\|_{L^q}$$

so taking the limit gives the result. ■

The case $p = 2$. For $p = 2$, $d_{L^p}(X_n, X)$ reduces to the *root mean-square error* between X_n and X converging to 0. Therefore, convergence in L^2 is also sometimes called mean-square convergence. Moreover, for $p = 2$, the L^2 space can be equipped with an inner product[7] $\langle X, Y\rangle_{L^2} := \mathbb{E}[X^\top Y]$, and the associated norm is $\langle X, X\rangle^{1/2} = \|X\|_{L^2}$. The space becomes a *Hilbert space*[8]. This matters a lot in statistics because inner products allow us to define notions of orthogonality, geometry, projections and therefore best approximations.

Exercise 1.5 (★).
Let $(X_n)_{n\in\mathbb{N}}$ and X be real random variables in L^2 such that $X_n \xrightarrow[n\to\infty]{L^2} X$. Show that

$$\lim_{n\to\infty} \mathbb{E}[X_n] = \mathbb{E}[X], \qquad \lim_{n\to\infty} \mathbb{E}[X_n^2] = \mathbb{E}[X^2], \qquad \lim_{n\to\infty} \mathbb{V}(X_n) = \mathbb{V}(X).$$

Hint: for the second limit, use the triangle inequality for the norm $\|\cdot\|_{L^2}$.

The projection theorem. Specifically, we say that two vectors x, y in a Hilbert space H are orthogonal, denoted $x \perp y$, if $\langle x, y\rangle = 0$. Let $M \subseteq H$ be a closed subspace (closed under addition, scalar multiplication, and for which limits of convergent sequences in M remain in M). The orthogonal complement of M is defined as

$$M^\perp = \{z \in H : \langle z, m\rangle = 0, \text{ for all } m \in M\},$$

[7] An inner product on a vector space V is a map $\langle\cdot,\cdot\rangle : V \times V \to \mathbb{R}$ satisfying the following three requirements: (1) Linearity in its first argument, $\langle ax + by, z\rangle = a\langle x, z\rangle + b\langle y, z\rangle$; (2) Symmetry, $\langle x, y\rangle = \langle y, x\rangle$; (3) Positive definiteness, $\langle x, x\rangle \geq 0$ and $\langle x, x\rangle = 0$ if and only if $x = 0$.

[8] A Hilbert space is an inner product space, complete under the norm induced by its inner product.

that is, the set of points orthogonal to all points of M. Then, every point $x \in H$ can be uniquely written as

$$x = P_M(x) + P_{M^\perp}(x), \qquad \text{with} \qquad P_M(x) \in M, \quad P_{M^\perp}(x) \in M^\perp.$$

The vector $P_M(x)$ is called the ***orthogonal projection*** of x onto M, and satisfies

$$\langle x - P_M(x), m \rangle = 0, \qquad \text{for all } m \in M.$$

In particular, it satisfies the ***best approximation property***: the projection $P_M(x)$ is the unique element of M closest to x:

$$\|x - P_M(x)\| \leqslant \|x - m\|, \qquad \text{for all } m \in M.$$

Exercise 1.6 (★★★).
Let $Y \in L^2$ be a real random variable, and let $H := L^2$ be equipped with the inner product $\langle X, Y \rangle_{L^2} = \mathbb{E}[XY]$.

(a) Let $M := \{c\mathbb{1}_\Omega : c \in \mathbb{R}\}$ be the subspace of constant random variables (it is closed, being finite-dimensional). Show that $P_M(Y) = \mathbb{E}[Y]$, and deduce that $\mathbb{E}[(Y-c)^2] \geq \mathbb{V}(Y)$ for every $c \in \mathbb{R}$, with equality if and only if $c = \mathbb{E}[Y]$.

(b) Let $X \in L^2$ with $\mathbb{V}(X) > 0$, and let $M := \{a + bX : a, b \in \mathbb{R}\}$, again a closed subspace. Show that

$$P_M(Y) = \mathbb{E}[Y] + \frac{\mathrm{Cov}(X,Y)}{\mathbb{V}(X)} \left(X - \mathbb{E}[X]\right),$$

where $\mathrm{Cov}(X,Y) := \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])]$. Deduce that the function $(a,b) \mapsto \mathbb{E}[(Y - a - bX)^2]$ is minimized at $b^* = \mathrm{Cov}(X,Y)/\mathbb{V}(X)$ and $a^* = \mathbb{E}[Y] - b^*\mathbb{E}[X]$.

(c) Show that the residual $R := Y - P_M(Y)$ of (b) satisfies $\mathbb{E}[R] = 0$ and $\mathrm{Cov}(X, R) = 0$.

1.1.5 Continuous mapping theorems and Slutsky's lemma

Continuous transformations. For a deterministic sequence $(x_n)_{n\in\mathbb{N}}$, continuity is exactly the concept needed to pass limit through functions, i.e., to write

$$\lim_{n\to\infty} x_n = x \qquad \Longrightarrow \qquad \lim_{n\to\infty} f(x_n) = f(\lim_{n\to\infty} x_n) = f(x).$$

The same idea essentially applies to stochastic convergence. We recall that $C(f)$ denotes the set of points at which f is continuous.

Theorem 1.8 (Continuous Mapping Theorem (CMT)).
Let f be measurable such that

$$\mathbb{P}[X \in C(f)] = 1.$$

Then,

$$X_n \xrightarrow[n\to\infty]{m} X \qquad \Longrightarrow \qquad f(X_n) \xrightarrow[n\to\infty]{m} f(X)$$

where the mode of convergence m is either *almost sure*, *in probability* or *in distribution*.

Proof. Proof of almost sure convergence. By almost sure convergence, there exists a set $\Omega_1 \subseteq \Omega$ on which X_n converges to X pointwise. On $\Omega_2 := X^{-1}(C(f))$, f is continuous. Hence, on their intersection, $f(X_n)$ converges pointwise to $f(X)$. We conclude by noting that the intersection of two almost sure events is an almost sure event.

Proof of convergence in probability. Let $(n_j)_{j\geq 1}$ be an increasing sequence in $\mathbb{N}$. Since X_n converges to X in probability, there exists a subsubsequence of $(X_{n_{j_k}})_{k\geq 1}$ that converges almost surely to X, and by the statement of the proof, $f(X_{n_{j_k}})$ converges almost surely to $f(X)$. Since $(n_{j_k})_{k\geq 1}$ was arbitrary, we have shown that every subsequence of $(f(X_n))_{n\in\mathbb{N}}$ has a further subsequence which converges almost surely, hence, $(f(X_n))_{n\in\mathbb{N}}$ converges in probability to $f(X)$.

Proof of convergence in distribution. Omitted, see Durrett (2019), Theorem 3.2.10. ■

It is worth noting that we do not need a function to be continuous everywhere for the CMT to apply. For example, if $X_n \xrightarrow[n\to\infty]{\mathbb{P}} 1$, then assuming that $\mathbb{P}[X_n = 0] = 0$, we have $1/X_n \xrightarrow[n\to\infty]{\mathbb{P}} 1$ since the map $x \mapsto 1/x$ is continuous at $x = 1$. Suppose $X_n \xrightarrow[n\to\infty]{m} X$, where m stands for either almost sure, in probability or in law. Then, for example,

$$||X_n|| \xrightarrow[n\to\infty]{m} ||X||, \qquad a^\top X_n \xrightarrow[n\to\infty]{m} a^\top X, \qquad X_n^\top A X_n \xrightarrow[n\to\infty]{m} X^\top A X,$$

for fixed vectors a and matrices A. Another interesting example is when $X_n \xrightarrow[n\to\infty]{\mathcal{L}} \mathcal{N}(0,1)$, then we directly know that $X_n^2 \xrightarrow[n\to\infty]{\mathcal{L}} \chi_1^2$.

This version of the CMT does not provide guarantees in general for continuous transformations in L^p. However, if we require that the function is L-Lipschitz[9], the result remains true.

Proposition 1.9 (CMT for L^p convergence)**.**
Let $p \geq 1$. Suppose that $X_n \xrightarrow[n\to\infty]{L^p} X$ and that f is Lipschitz. Then, $f(X_n) \xrightarrow[n\to\infty]{L^p} f(X)$.

Proof. Using the Lipschitz inequality, we get

$$\mathbb{E}\left[\|f(X_n) - f(X)\|^p\right] \leq L^p \mathbb{E}\left[\|X_n - X\|^p\right] \to 0$$

as $n \to \infty$. ■

1.1.6 Slutsky's theorem

With the continuous mapping theorem, we are equipped to look at transformations of sequences of random variables. However, suppose we are given $(X_n)_{n\in\mathbb{N}}$ and $(Y_n)_{n\in\mathbb{N}}$ such that $X_n \xrightarrow[n\to\infty]{m} X$ and $Y_n \xrightarrow[n\to\infty]{m} Y$, for some mode of convergence m, do we have the joint convergence $(X_n, Y_n) \xrightarrow[n\to\infty]{m} (X, Y)$? The answer is: it works for all modes of convergence, except for convergence in law.

The reason is that marginal distributions do not determine the dependence between the components. For example, let $Z \sim \mathcal{N}(0,1)$ and $X_n = Z$, $Y_n = (-1)^n Z$, $X = Z$ and $Y = Z$. Then, $X_n \xrightarrow[n\to\infty]{\mathcal{L}} X$ and $Y_n \xrightarrow[n\to\infty]{\mathcal{L}} Y$, but $(X_n, Y_n) = (Z, (-1)^n Z)$ does not converge to $(X, Y) = (Z, Z)$.

[9] A function f is Lipschitz with constant L if $\|f(x) - f(y)\| \leq L\|x - y\|$, for all $x, y \in \mathrm{dom}(f)$. This is a stronger notion than continuity: if a function is Lipschitz, it is uniformly continuous and thus continuous.

One exception, however, is when one of the two limiting objects X or Y is a constant. Then, the result does hold; this is the content of Slutsky's theorem.

Lemma 1.10 (Slutsky's lemma)**.**
Let c be a constant. If $X_n \xrightarrow[n\to\infty]{\mathcal{L}} X$ and $Y_n \xrightarrow[n\to\infty]{\mathcal{L}} c$, then

$$(X_n, Y_n) \xrightarrow[n\to\infty]{\mathcal{L}} (X, c)\,.$$

Proof. We prove the convergence using the Portmanteau theorem statement on Lipschitz functions. Let f be bounded Lipschitz with constant L. Then,

$$\left|\mathbb{E}\left[f(X_n, Y_n)\right] - \mathbb{E}\left[f(X, c)\right]\right| \leq \left|\mathbb{E}\left[f(X_n, Y_n)\right] - \mathbb{E}\left[f(X_n, c)\right]\right| + \left|\mathbb{E}\left[f(X_n, c)\right] - \mathbb{E}\left[f(X, c)\right]\right|,$$

by the triangle inequality. The second term converges to 0 since the map $x \mapsto f(x, c)$ is continuous and bounded. For the first, by Jensen's inequality

$$\begin{aligned}\left|\mathbb{E}\left[f(X_n, Y_n)\right] - \mathbb{E}\left[f(X_n, c)\right]\right| &\leq \mathbb{E}\left[\left|f(X_n, Y_n) - f(X_n, c)\right|\right] \\ &= \mathbb{E}\left[\left|f(X_n, Y_n) - f(X_n, c)\right|\mathbb{1}_{||Y_n - c|| \leq \epsilon}\right] + \mathbb{E}\left[\left|f(X_n, Y_n) - f(X_n, c)\right|\mathbb{1}_{||Y_n - c|| > \epsilon}\right]\end{aligned}$$

for an arbitrary $\epsilon > 0$. On the first part, using that f is Lipschitz,

$$\mathbb{E}\left[\left|f(X_n, Y_n) - f(X_n, c)\right|\mathbb{1}_{||Y_n - c|| \leq \epsilon}\right] \leq L\mathbb{E}\left[\left\|Y_n - c\right\|\mathbb{1}_{||Y_n - c|| \leq \epsilon}\right] \leq L\epsilon.$$

For the second term, we cannot control $||Y_n - c||$, however we know that $Y_n \xrightarrow[n\to\infty]{\mathbb{P}} c$ since convergence in distribution and in probability are equivalent when the limit is constant. Hence, using that f is bounded,

$$\begin{aligned}\mathbb{E}\left[\left|f(X_n, Y_n) - f(X_n, c)\right|\mathbb{1}_{||Y_n - c|| > \epsilon}\right] &\leq \mathbb{E}\left[\left(\left|f(X_n, Y_n)\right| + \left|f(X_n, c)\right|\right)\mathbb{1}_{||Y_n - c|| > \epsilon}\right] \\ &\leq 2\|f\|_\infty \mathbb{P}\left[||Y_n - c|| > \epsilon\right].\end{aligned}$$

This converges to zero and since ϵ was arbitrary, so does the second term. ∎

The idea that makes the result work is essentially that because Y_n concentrates around a constant, its dependence on X_n is asymptotically irrelevant. Slutsky's theorem is simply an application of the CMT and Slutsky's lemma.

Theorem 1.11 (Slutsky)**.**
Let c be a constant. Suppose that $X_n \xrightarrow[n\to\infty]{\mathcal{L}} X$ and $Y_n \xrightarrow[n\to\infty]{\mathcal{L}} c$. Then, for any function f such that

$$\mathbb{P}\left[(X, c) \in C(f)\right] = 1,$$

we have

$$f(X_n, Y_n) \xrightarrow[n\to\infty]{\mathcal{L}} f(X, c).$$

In particular, whenever the following operations are well-defined,

$$X_n + Y_n \xrightarrow[n\to\infty]{\mathcal{L}} X + c, \qquad X_n Y_n \xrightarrow[n\to\infty]{\mathcal{L}} cX, \qquad X_n Y_n^{-1} \xrightarrow[n\to\infty]{\mathcal{L}} X/c.$$

It is worth noting that the multiplication statement remains valid also for matrix-vector and matrix-matrix products; the ratio statement remains valid for matrices when the limiting matrix is non-singular.

1.2 Comparing modes of convergence: implications, failures, and upgrades

Previously, we defined the four most useful types of convergence for statistics. However, in practice, one often needs to navigate through them, or combine them in various ways. This is because the different types of convergence carry different types of information. Almost sure convergence follows the realized trajectories $\omega \to X_n(\omega)$; convergence in probability controls the probability of a large error at time n; convergence in L^p controls the average size of that error; convergence in distribution, however, only follows the laws of the random vectors. As briefly mentioned, some types of convergence are stronger than others, in the sense that they imply others. The goal of this section is to discuss the relationships between each type of convergence, which always imply which, and which can be upgraded to others, at the price of extra assumptions.

1.2.1 Basic implication diagram

The following theorem states the only implications which are always valid. Other implications do exist, but at the price of extra assumptions. Before proceeding, however, we introduce some very useful *concentration inequalities*. These will be used to prove the implication theorem, as well as many other times later on.

Concentration inequalities. A concentration inequality usually controls the deviation of a random variable from its mean. It often takes the form

$$\mathbb{P}\left[|X - \mathbb{E}[X]| > t\right] \leq f(t),$$

for some function of t, $f(t)$, that decreases as t increases. Here, we state two simple concentration inequalities, Markov and Chebyshev. Later on, we will introduce others too.

Lemma 1.12 (Markov's inequality)**.**
Let X be a nonnegative random variable. Then, for all $t > 0$

$$\mathbb{P}[X \geq t] \leq \frac{\mathbb{E}[X]}{t}.$$

Proof. Note that

$$X = X\mathbb{1}_{X \geq t} + X\mathbb{1}_{X < t} \geq t\mathbb{1}_{X \geq t},$$

since $X\mathbb{1}_{X \geq t} \geq t\mathbb{1}_{X \geq t}$ and $X\mathbb{1}_{X < t} \geq 0$ since $X \geq 0$. Taking expectation on both sides of the inequality gives

$$\mathbb{E}\left[X\right] \geq \mathbb{E}\left[t\mathbb{1}_{X \geq t}\right] = t\mathbb{E}\left[\mathbb{1}_{X \geq t}\right],$$

which, once reorganized, gives the inequality. ■

Markov's inequality is very useful because it will allow us to derive many more concentration inequalities. Our proof of Chebyshev's inequality will be based on it.

Lemma 1.13 (Chebyshev's inequality)**.**

Let X be a random variable such that $\mathbb{V}(X) < \infty$. Then, for all $t > 0$

$$\mathbb{P}\left[|X - \mathbb{E}[X]| \geq t\right] \leq \frac{\mathbb{V}(X)}{t^2}.$$

Proof. Since $x \mapsto x^2$ is an increasing function on $\mathbb{R}_+$, $\{|X - \mathbb{E}[X]| \geq t\} = \{|X - \mathbb{E}[X]|^2 \geq t^2\}$ so that

$$\mathbb{P}\left[|X - \mathbb{E}[X]| \geq t\right] = \mathbb{P}\left[|X - \mathbb{E}[X]|^2 \geq t^2\right] \leq \frac{\mathbb{E}\left[|X - \mathbb{E}[X]|^2\right]}{t^2} = \frac{\mathbb{V}(X)}{t^2},$$

where the above inequality follows by an application of Markov's inequality on the nonnegative random variable $|X - \mathbb{E}[X]|^2$. ■

We are now equipped to prove the following theorem.

Theorem 1.14 (Implications of different modes of stochastic convergence)**.**
The following implications hold.

(i) If $X_n \xrightarrow[n\to\infty]{a.s.} X$, then $X_n \xrightarrow[n\to\infty]{\mathbb{P}} X$.

(ii) If $X_n \xrightarrow[n\to\infty]{L^p} X$ for some $p \geq 1$, then $X_n \xrightarrow[n\to\infty]{\mathbb{P}} X$.

(iii) If $X_n \xrightarrow[n\to\infty]{\mathbb{P}} X$, then $X_n \xrightarrow[n\to\infty]{\mathcal{L}} X$.

Proof. Proof of (i). Assume that $X_n \xrightarrow[n\to\infty]{a.s.} X$. By Lemma 1.2, item (iv), this is equivalent to, for any $\epsilon > 0$,

$$\lim_{n\to\infty} \mathbb{P}\left[\sup_{k\geq n} \|X_k - X\| > \epsilon\right] = 0.$$

Since

$$\{\|X_n - X\| > \epsilon\} \subseteq \left\{\sup_{k\geq n} \|X_k - X\| > \epsilon\right\},$$

then

$$\mathbb{P}\left[\|X_n - X\| > \epsilon\right] \leq \mathbb{P}\left[\sup_{k\geq n} \|X_k - X\| > \epsilon\right] \tag{1.1}$$

by using the following fact.

Fact. A probability is a set increasing function, that is, if $A \subseteq B$, then $\mathbb{P}[A] \leq \mathbb{P}[B]$.

Taking the limit on both sides of (1.1) gives the result.

Proof of (ii). Let $\epsilon > 0$. By Chebyshev's/Markov's inequality on the nonnegative random variable $\|X_n - X\|^p$, we have

$$\mathbb{P}\left[\|X_n - X\| > \epsilon\right] \leq \frac{\mathbb{E}\left[\|X_n - X\|^p\right]}{\epsilon^p}.$$

Since ϵ is fixed, taking the limit on both sides gives the result.

Proof of (iii). This step is slightly more difficult; we follow the road of the Portmanteau theorem. Specifically, we wish to prove that, for any closed set F, $\limsup_{n\to\infty} \mathbb{P}[X_n \in F] \leq \mathbb{P}[X \in F]$.

However, by assuming that $X_n \xrightarrow[n\to\infty]{\mathbb{P}} X$, we only know that X_n is close to X with high probability, but this does not imply that $X \in F$ when X_n is. However, we can leverage that they are close.

Specifically, let F be closed, $\epsilon > 0$ and define $F^\epsilon := \{x \in \mathbb{R}^d : \text{dist}(x, F) \leq \epsilon\}$ to be the closed ϵ-enlargement of F. Notice that, if $X_n \in F$ and $\|X_n - X\| \leq \epsilon$, then $X \in F^\epsilon$.

> **Technique.** Suppose we wish to control the probability of an event A, and that we know of a high-probability event G on which things become easier. Then, we can decompose
>
> $$A = (A \cap G) \biguplus (A \cap G^c).$$
>
> Taking the probability on both sides gives
>
> $$\mathbb{P}[A] \leq \mathbb{P}[A \cap G] + \mathbb{P}[A \cap G^c] \leq \mathbb{P}[A \cap G] + \mathbb{P}[G^c].$$

We now leverage the above technique. We have a good event $\|X_n - X\| \leq \epsilon$ which holds with high probability, so

$$\mathbb{P}[X_n \in F] \leq \mathbb{P}\left[X_n \in F, \|X_n - X\| \leq \epsilon\right] + \mathbb{P}\left[\|X_n - X\| > \epsilon\right].$$

Since $X_n \in F$ and $\|X_n - X\| \leq \epsilon$ implies $X \in F^\epsilon$,

$$\mathbb{P}\left[X_n \in F, \|X_n - X\| \leq \epsilon\right] \leq \mathbb{P}[X \in F^\epsilon].$$

Hence,

$$\mathbb{P}[X_n \in F] \leq \mathbb{P}[X \in F^\epsilon] + \mathbb{P}\left[\|X_n - X\| > \epsilon\right].$$

Taking the limsup on both sides gives

$$\limsup_{n\to\infty} \mathbb{P}[X_n \in F] \leq \mathbb{P}[X \in F^\epsilon] + \lim_{n\to\infty} \mathbb{P}\left[\|X_n - X\| > \epsilon\right] = \mathbb{P}[X \in F^\epsilon].$$

In particular, since this holds for every $\epsilon > 0$, we have

$$\limsup_{n\to\infty} \mathbb{P}[X_n \in F] \leq \inf_{k\geq 1} \mathbb{P}[X \in F^{1/k}].$$

Notice that the sets $(F^{1/k})_{k\in\mathbb{N}^*} \downarrow \cap_{k\geq 1} F^{1/k} = \overline{F} = F$ since F is closed. Hence,

$$\inf_{k\geq 1} \mathbb{P}[X \in F^{1/k}] = \lim_{k\to\infty} \mathbb{P}[X \in F^{1/k}] = \mathbb{P}[X \in \lim_{k\to\infty} F^{1/k}] = \mathbb{P}[X \in F].$$

Overall

$$\limsup_{n\to\infty} \mathbb{P}[X_n \in F] \leq \mathbb{P}[X \in F],$$

which shows the result.

■

We summarize the "automatic" relationship between these modes of convergence in the following figure. We shall complete it soon, by adding how partial converses can be obtained (i.e., modes of convergence upgraded), at the price of additional assumptions.

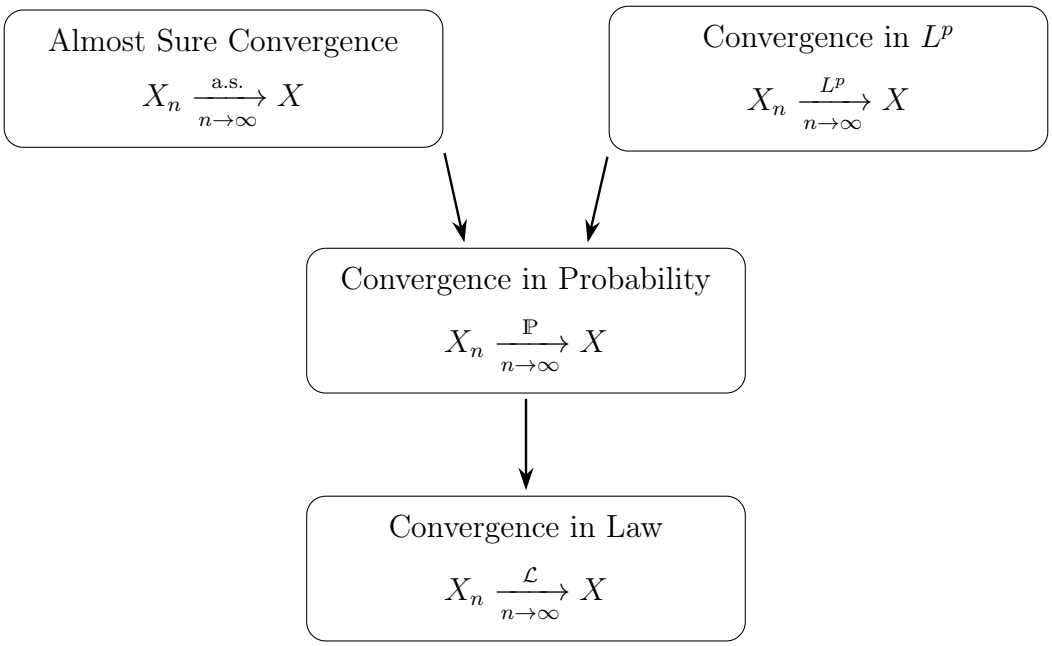


Figure 1.4. Relationships between several modes of convergence.

A first statistical application. Suppose an estimator $\widehat{\theta}_n \in \mathbb{R}$ estimates a parameter $\theta \in \mathbb{R}$, for which a computation gives

$$\mathbb{E}\left[|\widehat{\theta}_n - \theta|^2\right] \leq \frac{C}{n}$$

for some *universal constant*[10] C. From this, several things may be seen directly: as $n \to \infty$, $\mathbb{E}\left[|\widehat{\theta}_n - \theta|^2\right] \to 0$, which means that $\widehat{\theta}_n$ converges to θ in L^2. Therefore, convergence also holds in probability and in law, as per the previous diagram. Finally, since $\mathrm{MSE}(\widehat{\theta}_n) = \mathbb{E}\left[|\widehat{\theta}_n - \theta|^2\right] = \mathrm{Bias}^2(\widehat{\theta}_n) + \mathbb{V}(\widehat{\theta}_n)$, it follows that both $\mathrm{Bias}(\widehat{\theta}_n) \to 0$ and $\mathbb{V}(\widehat{\theta}_n) \to 0$ as $n \to \infty$.

1.2.2 Why converses fail

We briefly give a counterexample to all directions that do not appear in the previous diagram to show that, in fact, unless additional assumptions are made, it is not, in general, possible to upgrade these results.

Convergence in distribution does not imply convergence in probability. Let X be such that $\mathbb{P}[X = 1] = \mathbb{P}[X = -1] = 1/2$ and $X_n = -X$. By symmetry, $\mathcal{L}(X_n) = \mathcal{L}(-X) = \mathcal{L}(X)$ so that $X_n \xrightarrow[n\to\infty]{\mathcal{L}} X$. However, X_n and X never get close to each other: $|X_n - X| = 2$, a.s., for every n. The laws are the same, but the coupling between X_n and X makes them be always far away from each other.

[10] The term universal constant means a number that does not depend on any quantity that varies in the problem of interest. Here, for example, C may depend on many things, but not on n. Should we also consider the dimension d to be a variable of the model, C could not depend on it either.

Convergence in probability does not imply almost sure. Let $(X_n)_{n\in\mathbb{N}^*}$ be independent with law $\mathcal{B}(1/(n+1))$. Then, letting $X=0$, for any $\epsilon>0$, we have $\mathbb{P}[|X_n-X|>\epsilon]\to 0$ as $n\to\infty$. Indeed, if $\epsilon\geq 1$, then $\mathbb{P}[|X_n-X|>\epsilon]=0$ since for all n, $|X_n-X|\leq 1$ a.s. Now, for $\epsilon<1$,

$$\mathbb{P}\left[|X_n-X|>\epsilon\right]=\mathbb{P}\left[X_n=1\right]=\frac{1}{n+1}\to 0$$

as $n\to\infty$. This shows that $X_n \xrightarrow[n\to\infty]{\mathbb{P}} X$. However, X_n does not converge to 0 almost surely, because with probability 1, $X_n=1$ infinitely often. To see this, write

$$\mathbb{P}\left[X_n=1,\quad \text{i.o.}\right]=1-\mathbb{P}\left[X_n=1,\quad \text{f.o.}\right]=1-\mathbb{P}\left[X_n=0,\quad \text{a.a.}\right].$$

Now,

$$\begin{aligned}
\mathbb{P}\left[X_n=0,\quad \text{a.a.}\right] &= \mathbb{P}\left[\bigcup_{N\geq 1}\bigcap_{n\geq N}\{X_n=0\}\right]\\
&\leq \sum_{N\geq 1}\lim_{M\to\infty}\prod_{n=N}^{M}\mathbb{P}[X_n=0]\\
&= \sum_{N\geq 1}\lim_{M\to\infty}\prod_{n=N}^{M}\frac{n}{n+1}\\
&= \sum_{N\geq 1}\lim_{M\to\infty}\frac{N}{M+1}\\
&= 0.
\end{aligned}$$

This shows that $X_n=1$ infinitely often with probability one, and so X_n does not converge almost surely to 0. The underlying issue preventing almost sure convergence is the following: although at each sample size n, it is increasingly unlikely that $X_n=1$, there are infinitely many opportunities for it, so almost every sample path contains infinitely many of these events. Convergence in probability, on the other hand, only looks at one sample size n at a time, instead of looking at the whole path.

Almost sure convergence does not imply L^p. Let $\Omega=(0,1)$ with the Lebesgue measure and let

$$X_n(\omega)=n^{1/p}\mathbb{1}_{(0,1/n)}(\omega).$$

For any fixed $\omega>0$, $X_n(\omega)=0$ for all sufficiently large n; hence, $X_n \xrightarrow[n\to\infty]{a.s.} X$, with $X=0$. However,

$$\mathbb{E}\left[|X_n|^p\right]=n\mathbb{E}\left[\mathbb{1}_{(0,1/n)}\right]=n\mathbb{P}\left[(0,1/n)\right]=1,\qquad \text{for all } n\in\mathbb{N}^*.$$

Therefore, $\mathbb{E}\left[|X_n|^p\right]$ does not converge to 0, which shows that X_n does not converge to 0 in L^p. The reason for this "issue" is that almost sure convergence controls what happens at each fixed point ω, but does not control how large random variables can become on small probability sets. In contrast, L^p convergence is sensitive to both probability and magnitude. A rare event can still matter, even if the probability of this event is small.

A lack of information. Weaker modes of convergence do not generally imply their "stronger" counterparts because they do not carry enough information about the sequence $(X_n)_{n\in\mathbb{N}^*}$. Indeed:

- From convergence in law to probability, information about the coupling between X_n and X is missing.
- From probability to almost sure convergence, the missing information is how "bad events" are related along sample paths.
- From probability or almost sure to L^p, the missing information is the size of the variables on rare events.

1.2.3 Partial converse: convergence in distribution to a constant

There is one particular case where convergence in distribution does imply convergence in probability: when the limit random variable X is a constant.

Theorem 1.15 (Convergence in law to a constant)**.**
Let $c \in \mathbb{R}^d$ be a constant. Then, the following equivalence holds:

$$X_n \xrightarrow[n\to\infty]{\mathcal{L}} c \qquad \Longleftrightarrow \qquad X_n \xrightarrow[n\to\infty]{\mathbb{P}} c.$$

Proof. By Theorem 1.14, convergence in probability always implies in law; hence it is enough to show that, when the limit is constant, convergence in law does imply in probability.

Let $\epsilon > 0$ and define

$$F_\epsilon := \{x \in \mathbb{R}^d : \|x - c\| \geq \epsilon\}.$$

Notice that F_ϵ is closed, so that, by the Portmanteau lemma,

$$\limsup_{n\to\infty} \mathbb{P}\left[X_n \in F_\epsilon\right] \leq \mathbb{P}[X \in F_\epsilon] = 0$$

since $c \notin F_\epsilon$. Finally,

$$\mathbb{P}\left[X_n \in F_\epsilon\right] = \mathbb{P}\left[\|X_n - c\| \geq \epsilon\right] \geq \mathbb{P}\left[\|X_n - c\| > \epsilon\right]$$

which shows that $\lim_{n\to\infty} \mathbb{P}\left[\|X_n - c\| > \epsilon\right] = 0$ and thus convergence in probability holds. ■

The idea that makes the upgrade possible in the case of a constant lies in what information is missing by a convergence in law statement. In general, the laws of X_n and X do not say how they are coupled. However, for a constant $X = c$, there is no coupling choice to make, so that concentration of the law of X_n around δ_c, the law of X, is exactly concentration of X_n around c.

Exercise 1.7 (★★)**.**
Give a direct proof of Theorem 1.15 in the case $d = 1$, using only Definition 1.4: let $c \in \mathbb{R}$ and suppose that $X_n \xrightarrow[n\to\infty]{\mathcal{L}} c$.

(a) Identify F_c and $C(F_c)$.

(b) Show that, for every $\epsilon > 0$,

$$\mathbb{P}\left[|X_n - c| > \epsilon\right] \leq F_{X_n}(c - \epsilon) + 1 - F_{X_n}(c + \epsilon),$$

and conclude.

1.2.4 Partial converse: subsequences and almost sure convergence

Although convergence in probability does not force the complete sequence to converge almost surely, it does force almost sure convergence inside every subsequence in that every subsequence contains a further almost surely convergent subsequence.

Theorem 1.16 (Sub-subsequences and convergence in probability)**.**
The following statements are equivalent.

(i) $X_n \xrightarrow[n\to\infty]{\mathbb{P}} X$.

(ii) Every subsequence $(X_{n_k})_{k\in\mathbb{N}^*}$ contains a sub-subsequence $(X_{n_{k_j}})_{j\in\mathbb{N}^*}$ satisfying

$$X_{n_{k_j}} \xrightarrow[j\to\infty]{a.s.} X.$$

Proof. Proof of $(i) \implies (ii)$. Assume that $X_n \xrightarrow[n\to\infty]{\mathbb{P}} X$ and let $(X_{n_k})_{k\in\mathbb{N}^*}$ be an arbitrary subsequence. Since the whole sequence converges, $X_n \xrightarrow[n\to\infty]{\mathbb{P}} X$, then so does the extracted subsequence, $X_{n_k} \xrightarrow[k\to\infty]{\mathbb{P}} X$. We now extract a further subsequence. Let $k_0 = 0$ and, at step j, let $X_{n_{k_j}}$ satisfy $k_j > k_{j-1}$ and

$$\mathbb{P}\left[\|X_{n_{k_j}} - X\| > \frac{1}{j}\right] \leq 2^{-j}.$$

Let $A_j := \{\|X_{n_{k_j}} - X\| > j^{-1}\}$, for $j \in \mathbb{N}^*$. Then,

$$\begin{aligned}
\mathbb{P}\left[A_j, \text{ i.o.}\right] &= \mathbb{P}\left[\bigcap_{m\geq 1}\bigcup_{j\geq m} A_j\right] \\
&= \lim_{m\to\infty} \mathbb{P}\left[\bigcup_{j\geq m} A_j\right] \\
&\leq \lim_{m\to\infty} \sum_{j\geq m} \mathbb{P}\left[A_j\right] \\
&= \lim_{m\to\infty} \sum_{j\geq m} 2^{-j} \\
&= \lim_{m\to\infty} \frac{1}{2^{m-1}} \\
&= 0.
\end{aligned}$$

Thus, with probability one, only finitely many A_j occur. That is, with probability one,

$$\lim_{j\to\infty} \|X_{n_{k_j}}(\omega) - X(\omega)\| = 0$$

which shows almost sure convergence of that sub-subsequence.

Proof of $(ii) \implies (i)$. Suppose that every subsequence contains a further subsequence converging almost surely. We proceed by contradiction: assume that convergence in probability does not hold, i.e., there exists $\epsilon > 0$ such that $\mathbb{P}[\|X_n - X\| > \epsilon]$ does not converge to 0. Therefore, there exist $\rho > 0$ and infinitely many indices $n \in \mathbb{N}^*$ such that

$$\mathbb{P}\left[\|X_n - X\| > \epsilon\right] > \rho.$$

We can therefore construct a subsequence $(X_{n_k})_{k\geq 1}$ which satisfies, for every k,

$$\mathbb{P}\left[\|X_{n_k} - X\| > \epsilon\right] > \rho. \tag{1.2}$$

Since we assumed that every subsequence contains a further subsequence which converges almost surely, we can find $(X_{n_{k_j}})_{j\in\mathbb{N}^*}$ that converges almost surely to X, and thus, in probability as well. Hence,

$$\lim_{j\to\infty} \mathbb{P}\left[\|X_{n_{k_j}} - X\| > \epsilon\right] = 0.$$

However, this leads to a contradiction since we also have established that (1.2) holds for all $k \in \mathbb{N}^*$. ■

1.2.5 Uniform integrability and upgrades

Limits and expectations. As illustrated in the counterexample in Section 1.2.2, some rarer and rarer bad events may have a non-negligible amount of expected mass, hence preventing convergence in L^p. In general, we may be interested in the question of "interchanging" limits and expectations, namely, whether the operation

$$\lim_{n\to\infty} \mathbb{E}[X_n] = \mathbb{E}\left[\lim_{n\to\infty} X_n\right],$$

and related operations. To discuss this in more detail, three tools are commonly introduced: the monotone convergence theorem, Fatou's lemma, and the dominated convergence theorem.

Monotone Convergence Theorem (MCT): Suppose $X_n \geq 0$ and $X_1 \leq X_2 \leq \ldots$ is increasing, with $X_n \xrightarrow[n\to\infty]{a.s.} X$. Then,

$$\lim_{n\to\infty} \mathbb{E}[X_n] = \mathbb{E}\left[\lim_{n\to\infty} X_n\right] = \mathbb{E}[X].$$

Note that this does not require integrability of X, the limit could be $+\infty$. More generally, the conclusion of the theorem works if $X_n \geq Z$ for some $Z \in L^1$ and $X_n \uparrow X$ a.s., or if $X_n \leq Z$ for some $Z \in L^1$ and $X_n \downarrow X$.

Fatou's lemma: If $X_n \geq 0$, then

$$\mathbb{E}\left[\liminf_{n\to\infty} X_n\right] \leq \liminf_{n\to\infty} \mathbb{E}\left[X_n\right].$$

Note that, technically, we do not need $X_n \geq 0$, but rather that $X_n \geq Z$ for some random variable $Z \in L^1$. Note that strict inequality does hold sometimes: let $X_n = n\mathbb{1}_{U\leq 1/n}$, with $U \sim \mathcal{U}([0,1])$. Then, we have

$$\mathbb{P}\left[\lim_{n\to\infty} X_n \neq 0\right] = \mathbb{P}\left[U = 0\right] = 0.$$

Hence, the set of points at which X_n converges to 0 has probability one, so that $X_n \xrightarrow[n\to\infty]{a.s.} 0$. Hence, almost surely, $\liminf_{n\to\infty} X_n = \lim_{n\to\infty} X_n$ so that Fatou's lemma gives

$$0 = \mathbb{E}[0] = \mathbb{E}\left[\liminf_{n\to\infty} X_n\right] \leq \liminf_{n\to\infty} \mathbb{E}[X_n] = 1.$$

Dominated Convergence Theorem (DCT): If $|X_n| \leq Y$ with $Y \in L^1$, and $X_n \xrightarrow[n\to\infty]{a.s.} X$, then $X_n \xrightarrow[n\to\infty]{L^1} X$. It implies what is sometimes called the ***bounded convergence theorem***, which states that if $|X_n| \leq C$ for some constant C, and $X_n \xrightarrow[n\to\infty]{a.s.} X$, then $X_n \xrightarrow[n\to\infty]{L^1} X$. This is direct since C is itself an integrable random variable. However, although these upgrades are valid and often useful, they require conditions that are stronger than necessary. The right concept is that of *uniform integrability*.

Definition 1.17 (Uniform integrability).
A sequence of integrable random variables $(X_n)_{n\in\mathbb{N}^*}$ is ***uniformly integrable (UI)*** if

$$\lim_{M\to\infty} \sup_{n\geq 1} \mathbb{E}\left[\|X_n\| \mathbb{1}_{\|X_n\|\geq M}\right] = 0.$$

Uniform integrability can be understood through the idea of uniformly approximating random variables by bounded versions, arbitrarily well. For $X \in L^1$, define its M-bounded approximation

$$X^{(M)} = X\mathbb{1}_{\|X\|<M}.$$

It holds that

$$X = X\mathbb{1}_{\|X\|<M} + X\mathbb{1}_{\|X\|\geq M} = X^{(M)} + X\mathbb{1}_{\|X\|\geq M}.$$

Hence, we can view the norm of the random variable $X\mathbb{1}_{\|X\|\geq M}$ as an approximation error:

$$\|X\mathbb{1}_{\|X\|\geq M}\|_{L^1} = \|X - X^{(M)}\|_{L^1}.$$

Now, consider the following statement.

Exercise 1.8 (★★). Show that, if $X \in L^1$,

$$\lim_{M\to\infty} \|X - X^{(M)}\|_{L^1} = 0.$$

Hence, it holds that, for any fixed n,

$$\lim_{M\to\infty} \|X_n - X_n^{(M)}\|_{L^1} = 0.$$

In other words, for every $n \in \mathbb{N}$, for every $\epsilon > 0$, there exists $M(\epsilon, n)$ such that

$$\|X_n - X_n^{(M(\epsilon,n))}\|_{L^1} < \epsilon.$$

That is, it is possible to approximate as well as one wishes any integrable random variable by a bounded random variable. The question remaining, however, happens when we consider a sequence of random variables $(X_n)_{n\in\mathbb{N}}$: can we approximate a sequence of integrable random variables $(X_n)_{n\in\mathbb{N}}$ by the uniformly bounded sequence $(X_n^{(M)})_{n\in\mathbb{N}}$, arbitrarily well? The key here is: uniformly bounded. In the above example, we could, for any n, find a cutoff $M(\epsilon, n)$ that could

make the approximation error as small as possible. The issue here is that $M(\epsilon, n)$ depends on n. We thus wish to know if we can find a cutoff $M = M(\epsilon)$, that works uniformly over all $n \in \mathbb{N}$. If we can find one, the sequence is uniformly integrable; otherwise, it is not.

A first useful observation is that uniform integrability permits a uniform control of moments. If, moreover, one has convergence in probability, this can be upgraded to show that the probability limit itself is integrable. To prove this, we will make use of Fatou's lemma.

Lemma 1.18 (UI and integrability)**.**
Suppose that $(X_n)_{n\in\mathbb{N}}$ is UI. Then, the following holds.

(i) $\sup_{n\in\mathbb{N}} \mathbb{E}[\|X_n\|] < \infty$.

(ii) If $X_n \xrightarrow[n\to\infty]{\mathbb{P}} X$, then $\mathbb{E}[\|X\|] < \infty$.

Proof. Proof of (i). Let $M > 0$. Then,

$$\sup_{n\in\mathbb{N}} \mathbb{E}\left[\|X_n\|\right] \leq \sup_{n\in\mathbb{N}} \mathbb{E}\left[\|X_n\|\mathbb{1}_{\|X_n\|\leq M}\right] + \sup_{n\in\mathbb{N}} \mathbb{E}\left[\|X_n\|\mathbb{1}_{\|X_n\|>M}\right] \leq M + \sup_{n\in\mathbb{N}} \mathbb{E}\left[\|X_n\|\mathbb{1}_{\|X_n\|>M}\right].$$

By UI, since $\sup_{n\in\mathbb{N}} \mathbb{E}\left[\|X_n\|\mathbb{1}_{\|X_n\|>M}\right] \to 0$ as $M \to \infty$, we can find M such that $\sup_{n\in\mathbb{N}} \mathbb{E}\left[\|X_n\|\mathbb{1}_{\|X_n\|>M}\right] < \epsilon$ for some $\epsilon > 0$. Then,

$$\sup_{n\in\mathbb{N}} \mathbb{E}\left[\|X_n\|\right] \leq M + \epsilon < \infty.$$

Proof of (ii). Since $X_n \xrightarrow[n\to\infty]{\mathbb{P}} X$, by the continuous mapping theorem, $\|X_n\| \xrightarrow[n\to\infty]{\mathbb{P}} \|X\|$. Hence, there exists a subsequence $(X_{n_k})_{k\in\mathbb{N}}$ such that $\|X_{n_k}\| \xrightarrow[k\to\infty]{a.s.} \|X\|$. For this sequence, we have equality of the random variables $\liminf_{k\to\infty} \|X_{n_k}\| = \lim_{k\to\infty} \|X_{n_k}\| = \|X\|$. By Fatou's lemma,

$$\mathbb{E}\left[\|X\|\right] = \mathbb{E}\left[\liminf_{k\to\infty} \|X_{n_k}\|\right] \leq \liminf_{k\to\infty} \mathbb{E}[\|X_{n_k}\|] \leq \sup_{n\in\mathbb{N}} \mathbb{E}[\|X_n\|] < \infty$$

by item (i).

■

Lemma 1.19.
Assume that $(X_n)_{n\in\mathbb{N}}$ satisfies the following two hypotheses.

(H1) $X_n \xrightarrow[n\to\infty]{\mathbb{P}} 0$.

(H2) $(X_n)_{n\in\mathbb{N}}$ is UI.

Then,

$$\lim_{n\to\infty} \mathbb{E}[\|X_n\|] = 0.$$

Proof. We split $\mathbb{E}[\|X_n\|]$ into three regimes: small values, moderate values and large values. Small values will be easy to control, moderate values will be controlled using convergence in probability, and large values with UI.

Let $\epsilon > 0$, $\delta > 0$, and $M > \epsilon$ such that $\sup_{n\in\mathbb{N}} \mathbb{E}\left[\|X_n\|\mathbb{1}_{\|X_n\|\geq M}\right] < \delta$. Write

$$\mathbb{E}[\|X_n\|] = \mathbb{E}\left[\|X_n\|\mathbb{1}_{\|X_n\|\leq\epsilon}\right] + \mathbb{E}\left[\|X_n\|\mathbb{1}_{\epsilon<\|X_n\|<M}\right] + \mathbb{E}\left[\|X_n\|\mathbb{1}_{\|X_n\|\geq M}\right] \leq \epsilon + M\mathbb{P}(\|X_n\| > \epsilon) + \delta.$$

By letting $n \to \infty$, we have $\mathbb{P}(\|X_n\| > \epsilon) \to 0$. Since ϵ and δ were arbitrary, taking $\epsilon \to 0$ and $\delta \to 0$ gives the result. ■

The above lemma will help us provide a simple proof of ***Vitali's convergence theorem***, providing us with the tool to upgrade convergence in probability to convergence in L^p. The following exercise will also be useful.

Exercise 1.9 (★★★).
Let $p \geq 1$ and suppose that $X \in L^p$. Then, for any sequence $(X_n)_{n\in\mathbb{N}}$, the following equivalence holds:

$$(||X_n - X||^p)_{n\in\mathbb{N}} \text{ is UI} \qquad \Longleftrightarrow \qquad (||X_n||^p)_{n\in\mathbb{N}} \text{ is UI.}$$

Note that the result of the above exercise does not require X_n to converge to X.

Theorem 1.20 (Vitali's convergence theorem).
Let $p \geq 1$, and suppose that $X_n \xrightarrow[n\to\infty]{\mathbb{P}} X$. Then, the following statements are equivalent.

(i) $X_n \xrightarrow[n\to\infty]{L^p} X$.

(ii) $(||X_n||^p)_{n\in\mathbb{N}}$ is UI.

Proof. Proof of (i) ⇒ (ii). Since $\mathbb{E}\left[||X_n - X||^p\right] \to 0$ as $n \to \infty$, there exists $N \in \mathbb{N}$ such that for all $n \geq N$, $\mathbb{E}\left[||X_n - X||^p\right] < \epsilon$. Thus, for $M > 0$ and $n \geq N$,

$$\mathbb{E}\left[||X_n - X||^p\mathbb{1}_{||X_n-X||>M}\right] \leq \mathbb{E}\left[||X_n - X||^p\right] < \epsilon.$$

By an extension of Exercise 1.8 to the case of approximating finitely many random variables, we can pick M large enough so that

$$\max_{1\leq n\leq N} \mathbb{E}\left[||X_n - X||^p\mathbb{1}_{||X_n-X||>M}\right] \leq \epsilon.$$

Overall, for any $\epsilon > 0$, there exists M such that

$$\sup_{n\in\mathbb{N}} \mathbb{E}\left[||X - X_n||^p\mathbb{1}_{||X_n-X||>M}\right] \leq \epsilon.$$

Since $X \in L^p$, an application of Exercise 1.9 gives the result.

Proof of (ii) ⇒ (i). Since $X_n \xrightarrow[n\to\infty]{\mathbb{P}} X$, we have $\|X_n - X\| \xrightarrow[n\to\infty]{\mathbb{P}} 0$. By continuity of $x \mapsto x^p$, we have $\|X_n - X\|^p \xrightarrow[n\to\infty]{\mathbb{P}} 0$. Hence, by (ii), we have uniform integrability of $||X_n||^p$, and by Exercise 1.9, provided that $X \in L^p$, it follows that $(||X_n - X||^p)_{n\in\mathbb{N}}$ is UI. Thus, an application of Lemma 1.19 would give the conclusion. The fact that $X \in L^p$ follows by Lemma 1.18.

■

Sufficient conditions for uniform integrability

In practice, one rarely verifies uniform integrability by its definition. A few other facts imply uniform integrability; these are often simpler to work with.

Proposition 1.21 (Criteria to verify UI)**.**
Let $(X_n)_{n\in\mathbb{N}}$ be integrable. Then, the following statements hold.

(i) If $||X_n|| < Y$ for every $n \in \mathbb{N}$ with $Y \in L^1$, then $(X_n)_{n\in\mathbb{N}}$ is UI.

(ii) If there exists $r > 1$ such that

$$\sup_{n\in\mathbb{N}} \mathbb{E}[||X_n||^r] < \infty,$$

then $(X_n)_{n\in\mathbb{N}}$ is UI.

Proof. Proof of (i). Since $||X_n|| < Y$, it holds that $\{||X_n|| > M\} \subset \{Y > M\}$ and thus

$$\mathbb{E}\left[||X_n||\mathbb{1}_{||X_n||>M}\right] \leq \mathbb{E}\left[Y\mathbb{1}_{||X_n||>M}\right] \leq \mathbb{E}\left[Y\mathbb{1}_{Y>M}\right].$$

Since the quantity on the far right-hand side does not depend on n, UI follows.

Proof of (ii). On the event $\{||X_n|| > M\}$, since $r > 1$, the map $x \mapsto x^r$ is increasing so that

$$||X_n|| > M \qquad \Longrightarrow \qquad ||X_n||^{r-1} > M^{r-1}.$$

Hence,

$$||X_n|| = \frac{||X_n||^r}{||X_n||^{r-1}} \leq \frac{||X_n||^r}{M^{r-1}}.$$

Hence,

$$\sup_{n\in\mathbb{N}} \mathbb{E}\left[||X_n||\mathbb{1}_{||X_n||>M}\right] \leq \frac{\sup_{n\in\mathbb{N}} \mathbb{E}\left[||X_n||^r\right]}{M^{r-1}}$$

which converges to 0 as $M \to \infty$. ■

It is interesting to note that UI implies $\sup_{n\in\mathbb{N}} \mathbb{E}[\|X_n\|] < \infty$. However, the converse is not true, UI is rather implied by $\sup_{n\in\mathbb{N}} \mathbb{E}[\|X_n\|^{1+\delta}] < \infty$ for some $\delta > 0$.

Exercise 1.10 (★★)**.**
Let $U \sim \mathcal{U}([0,1])$ and $X_n := n\mathbb{1}_{\{U\leq 1/n\}}$, the sequence used to illustrate Fatou's lemma.

(a) Show that $\sup_{n\in\mathbb{N}^*} \mathbb{E}[X_n] = 1 < \infty$ and that $X_n \xrightarrow[n\to\infty]{a.s.} 0$.

(b) Show that $(X_n)_{n\in\mathbb{N}^*}$ is not UI.

(c) Deduce from Theorem 1.20, without any computation, that X_n does not converge to 0 in L^1. Confirm by a direct computation.

(d) Show that $\sup_{n\in\mathbb{N}^*} \mathbb{E}[X_n^r] = \infty$ for every $r > 1$, in accordance with Proposition 1.21 (ii).

Some examples. Suppose we know that $X_n \xrightarrow[n\to\infty]{\mathbb{P}} X$ and, for all $n \in \mathbb{N}$, $\|X_n\| \leq C$, for some constant $C > 0$. Then, for every fixed $p \in \mathbb{N}$, $(||X_n||^p)_{n\in\mathbb{N}}$ is UI since $||X_n||^p \leq C^p$. Hence, $X_n \xrightarrow[n\to\infty]{L^p} X$.

Suppose now that $X_n \xrightarrow[n\to\infty]{\mathbb{P}} X$ but only $\sup_{n\in\mathbb{N}} \mathbb{E}[\|X_n\|^2] < \infty$. Then, $(||X_n||)_{n\in\mathbb{N}}$ is UI so $X_n \xrightarrow[n\to\infty]{L^1} X$. However, we a priori do not know that $(||X_n||^2)_{n\in\mathbb{N}}$ is UI, since this would require knowing, for example, that $\sup_{n\in\mathbb{N}} \mathbb{E}[\|X_n\|^{2+\delta}] < \infty$ for some $\delta > 0$.

Convergence in distribution and uniform integrability

Convergence in distribution, by the Portmanteau theorem, guarantees convergence of bounded continuous functions. That is, if $X_n \xrightarrow[n\to\infty]{\mathcal{L}} X$ then $\mathbb{E}[f(X_n)] \to \mathbb{E}[f(X)]$ as $n \to \infty$ for any f both continuous and bounded. Moments $x \mapsto x^q$ are continuous, but not bounded functions; hence, a priori, convergence in distribution does not provide us with convergence of the type $\mathbb{E}[X_n^2] \to \mathbb{E}[X^2]$ as $n \to \infty$, for example. The key missing ingredient to obtain convergence of moments from convergence in distribution is again uniform integrability. Moments of course have nothing special about that statement, we can essentially remove the bounded requirement provided that the sequence is UI.

Proposition 1.22 (Law + UI = Convergence of expectations for continuous functions)**.** Suppose $X_n \xrightarrow[n\to\infty]{\mathcal{L}} X$. Assume that:

(H1) f is a continuous function.

(H2) The sequence $(f(X_n))_{n\in\mathbb{N}}$ is UI.

Then,

$$\lim_{n\to\infty} \mathbb{E}[f(X_n)] = \mathbb{E}[f(X)].$$

Proof. We prove the result when f is positive. The extension to signed f is left as an exercise.

Let $Y_n := f(X_n)$, with $Y := f(X)$ and consider the truncation $t_M : x \mapsto \min(x, M)$ for some $M > 0$. Then, consider the decomposition

$$\big|\mathbb{E}[Y_n] - \mathbb{E}[Y]\big| \leq \big|\mathbb{E}[Y_n] - \mathbb{E}[t_M(Y_n)]\big| + \big|\mathbb{E}[t_M(Y_n)] - \mathbb{E}[t_M(Y)]\big| + \big|\mathbb{E}[t_M(Y)] - \mathbb{E}[Y]\big|.$$

We will show that each term converges to zero. The decomposition is valid for all $M > 0$, we choose one such that $\mathbb{E}[Y_n \mathbb{1}_{Y_n>M}] \leq \epsilon$ for all n. Then, we note that

$$\mathbb{E}\left[Y_n - t_M(Y_n)\right] = \mathbb{E}\left[\mathbb{1}_{Y_n>M}\left(Y_n - M\right)\right] \leq \mathbb{E}\left[\mathbb{1}_{Y_n>M} Y_n\right] \leq \epsilon.$$

For the second term, we note that t_M is a continuous bounded function: by the Portmanteau theorem, this term converges to 0. Finally, for the last term, we can verify that $t_M(Y)$ converges almost surely to Y as $M \to \infty$ and,

$$Y - t_M(Y) = \mathbb{1}_{Y>M}\left(Y - M\right) \leq \mathbb{1}_{Y>M} Y \leq Y$$

and $Y \in L^1$ so that an application of the DCT gives the result. To see why $Y \in L^1$, note that

$$\mathbb{E}[t_M(Y)] = \lim_{n\to\infty} \mathbb{E}[t_M(Y_n)] \leq \sup_{n\in\mathbb{N}} \mathbb{E}[Y_n] < \infty$$

by the Portmanteau theorem and UI. Finally, letting $M \to \infty$ and the MCT gives

$$\lim_{M\to\infty} \mathbb{E}[t_M(Y)] = \mathbb{E}[Y] < \infty.$$

■

Consistency, unbiasedness. Suppose we know that $\sqrt{n}(\widehat{\theta}_n - \theta) \xrightarrow[n\to\infty]{\mathcal{L}} \mathcal{N}(0,1)$ and that

$$\sup_{n\in\mathbb{N}} \mathbb{E}\left[\left|\sqrt{n}(\widehat{\theta}_n - \theta)\right|^{2+\delta}\right] < \infty.$$

By just knowing the limiting distribution of $\widehat{\theta}_n$, we cannot certify that $\lim_{n\to\infty} \mathbb{E}[\widehat{\theta}_n] = \theta$. In other words, although we know that $\widehat{\theta}_n \xrightarrow[n\to\infty]{\mathbb{P}} \theta$, meaning in statistical terms that $\widehat{\theta}_n$ is consistent, we cannot certify that it is asymptotically unbiased. In other words, consistency does not in general imply asymptotic unbiasedness; obviously, asymptotic unbiasedness does not prove consistency either. Using the uniform moment bound, however, we get directly that $(\sqrt{n}(\widehat{\theta}_n - \theta))_{n\in\mathbb{N}}$ and $(n(\widehat{\theta}_n - \theta)^2)_{n\in\mathbb{N}}$ are UI, so that indeed

$$\mathbb{E}\left[\widehat{\theta}_n\right] - \theta = o\left(\frac{1}{\sqrt{n}}\right), \qquad \text{and} \qquad \lim_{n\to\infty} n\mathbb{E}\left[\left(\widehat{\theta}_n - \theta\right)^2\right] = 1.$$

Together, these statements can be combined to show that

$$\lim_{n\to\infty} n\mathbb{V}(\widehat{\theta}_n) = 1.$$

Exercise 1.11 (★★★).
Let $X_1, ..., X_n$ be i.i.d. real random variables with mean θ and variance 1, and admit that $\sqrt{n}\left(\overline{X}_n - \theta\right) \xrightarrow[n\to\infty]{\mathcal{L}} Z \sim \mathcal{N}(0,1)$. Let $U \sim \mathcal{U}([0,1])$ be independent of the sample, let $A_n := \{U \leq 1/n^2\}$, and define the estimator

$$\widehat{\theta}_n := \overline{X}_n + n^2 \mathbb{1}_{A_n}.$$

(a) Show that $\widehat{\theta}_n \xrightarrow[n\to\infty]{\mathbb{P}} \theta$: the estimator is consistent.

(b) Show that $\sqrt{n}\left(\widehat{\theta}_n - \theta\right) \xrightarrow[n\to\infty]{\mathcal{L}} \mathcal{N}(0,1)$: the estimator is asymptotically normal, with the same limit as $\overline{X}_n$.

(c) Show that $\mathbb{E}[\widehat{\theta}_n] = \theta + 1$ for every n: the bias does not vanish.

(d) Let $W_n := \sqrt{n}\left(\widehat{\theta}_n - \theta\right)$. Using Proposition 1.22, show that $(W_n)_{n\in\mathbb{N}}$ is not UI, and deduce that $\sup_{n\in\mathbb{N}} \mathbb{E}[|W_n|^{1+\delta}] = \infty$ for every $\delta > 0$.

1.2.6 Summability and upgrade to almost sure convergence

Convergence in probability says that the probabilities of large errors tend to zero. To obtain almost sure convergence, it is enough to require them to tend to zero "sufficiently fast". Indeed, almost sure convergence requires that, for all $\epsilon > 0$, $\mathbb{P}\left[\|X_n - X\| > \epsilon \text{ i.o.}\right] = 0$. That is, the

number of times the event $A_n = \{\|X_n - X\| > \epsilon\}$ happens must be finite with probability one, for almost sure convergence to hold. As a tool, let $N := \sum_{n\geq 1} \mathbb{1}_{A_n}$ count the number of bad events occurring.

The expected number of bad events is given by

$$\mathbb{E}[N] = \sum_{n\geq 1} \mathbb{P}[A_n].$$

To see this, recall that two theorems are frequently used to justify changing the order of integration: Fubini and Tonelli. Both lead to the same conclusion, but under different assumptions.

Let (X, μ) and (Y, ν) be two measure spaces[11] and $f : X \times Y \to \mathbb{R}$. Suppose that either:

- ***Tonelli's theorem***: The function is nonnegative, i.e., $f(x, y) \geq 0$.
- ***Fubini's theorem***: The function is integrable:

$$\int_{X\times Y} |f(x, y)|(\mu \otimes \nu)(\mathrm{d}x, \mathrm{d}y) < \infty.$$

Then,

$$\int_{X\times Y} f(x, y)(\mu \otimes \nu)(\mathrm{d}x, \mathrm{d}y) = \int_X \left(\int_Y f(x, y)\nu(\mathrm{d}y)\right) \mu(\mathrm{d}x) = \int_Y \left(\int_X f(x, y)\mu(\mathrm{d}x)\right) \nu(\mathrm{d}y).$$

The key observation is that Tonelli does not require finiteness of the joint integral, only positivity. Therefore, in practice, if we wish to interchange the order of integration of a positive function, then Tonelli gives the result. When we wish to do so to a potentially negative function, we use Tonelli on its absolute value to show that the integral is finite, then we can apply Fubini.

Coming back to our discussion on the upgrade from convergence in probability to almost sure, we may use the following technique.

> **Technique.** To interchange a series and an expectation, we may view the series as an integral with respect to the counting measure.

In our case, the function is simply an indicator function, $\mathbb{1}_{A_n}$, hence positive. Thus,

$$\mathbb{E}\left[\sum_{n\geq 1} \mathbb{1}_{A_n}\right] = \sum_{n\geq 1} \mathbb{E}\left[\mathbb{1}_{A_n}\right] = \sum_{n\geq 1} \mathbb{P}[A_n].$$

Overall, we see that if $\sum_{n\geq 1} \mathbb{P}[A_n] < \infty$, then $\mathbb{E}[N] < \infty$, which means that N is almost surely finite. The same technique applies to any sequence of events $(A_n)_{n\in\mathbb{N}}$ that one wishes to show occur only finitely many times.

> **Lemma 1.23** (Borel-Cantelli Lemma).
> Let $(A_n)_{n\in\mathbb{N}}$ be a sequence of events. If
>
> $$\sum_{n\geq 1} \mathbb{P}[A_n] < \infty,$$

[11] Technically, we need these measure spaces to be σ-finite; this is trivially satisfied in this course with probability spaces.

then $\mathbb{P}[A_n, \text{ i.o.}] = 0$.

Theorem 1.24 (Summability criterion for a.s. convergence).
Suppose that, for all $\epsilon > 0$,

$$\sum_{n \geq 1} \mathbb{P}\left[\|X_n - X\| > \epsilon\right] < \infty.$$

Then, $X_n \xrightarrow[n\to\infty]{a.s.} X$.

1.3 Stochastic asymptotic notation and rates

Asymptotic statements are essentially approximations done by neglecting negligible terms here and there. This holds for asymptotics in probability and statistics, but also in other areas of mathematics. We will use a set of symbols $\mathcal{O}, o, \sim, \asymp, \lesssim, \gtrsim$ that will allow us to greatly simplify our arguments, proof lengths, and results. We start by discussing their definitions in a deterministic setting before extending to their probabilistic counterparts.

1.3.1 Deterministic $\mathcal{O}, o, \sim, \asymp, \lesssim, \gtrsim$

In the sequel, we let $(a_n)_{n\in\mathbb{N}}$ be a positive deterministic sequence and $(x_n)_{n\in\mathbb{N}}$ be a sequence in $\mathbb{R}^d$.

Definition 1.25 (Order symbols).

- We write $x_n = \mathcal{O}(a_n)$ if there exist $C < \infty$ and $N \in \mathbb{N}$ such that for all $n \geq N$,

$$\|x_n\| \leq C a_n.$$

- We write $x_n = o(a_n)$ if

$$\lim_{n\to\infty} \frac{\|x_n\|}{a_n} = 0.$$

Restricting our attention to positive sequences, we also write

$$\begin{aligned}
a_n \sim b_n \quad &\Leftrightarrow \quad \lim_{n\to\infty} \frac{a_n}{b_n} = 1. \\
a_n \lesssim b_n \quad &\Leftrightarrow \quad a_n = \mathcal{O}(b_n). \\
a_n \gtrsim b_n \quad &\Leftrightarrow \quad b_n = \mathcal{O}(a_n). \\
a_n \asymp b_n \quad &\Leftrightarrow \quad a_n \lesssim b_n \text{ and } a_n \gtrsim b_n.
\end{aligned}$$

We also write $x_n = \mathcal{O}(1)$ if it is a bounded sequence, and we have $x_n = \mathcal{O}(a_n)$ if and only if $x_n/a_n = \mathcal{O}(1)$.

These symbols being very useful, it is worth trying to get some intuition.

- The notation $x_n = \mathcal{O}(a_n)$ means that x_n is of order smaller than or equal to a_n.

- The notation $x_n = o(a_n)$ means that x_n is negligible compared to a_n.
- The notation $a_n \sim b_n$ means a_n and b_n are asymptotically equivalent.
- The notation $a_n \asymp b_n$ means that a_n and b_n are of the same order, which can also be written as: there exist constants $0 < C < C' < \infty$ such that, for all sufficiently large n,

$$C \leq \frac{a_n}{b_n} \leq C'.$$

  That is, the ratio of the two sequences remains bounded away from zero and infinity.

For example, $n^2 + 3n \sim n^2$, $2n \asymp n$, and $\log(n) = o(n^\alpha), \alpha > 0$. More generally, consider the sequence

$$a_n = \frac{a}{n} + \frac{b}{n\sqrt{n}} + \frac{c}{n^2} + \frac{d}{n^2\sqrt{n}}, \qquad n \in \mathbb{N}.$$

Then,

- $a_n \sim n^{-2}$ if and only if $a = 0, b = 0, c = 1$.
- $a_n \asymp n^{-2}$ if and only if $a = 0$, $b = 0$, $c \neq 0$.
- $a_n = \mathcal{O}(n^{-2})$ if and only if $a = 0$, $b = 0$.
- $a_n = o(n^{-2})$ if and only if $a = 0$, $b = 0$, $c = 0$.

The meaning of $a_n = o(b_n)$ slightly depends on the behaviors of the sequences: when both tend to infinity, it means that b_n tends to infinity faster than a_n; when they tend to zero, it means a_n tends to zero faster than b_n. For example, $n = o(n^2)$, and $1/n^2 = o(1/n)$. In both cases, the quantity on the left is small compared to that on the right.

In $x_n = \mathcal{O}(a_n)$, it is also worth noting that the requirement $n \geq N$ is equivalent to that of requiring it for all $n \in \mathbb{N}$. Indeed, if for all $n \geq N$, $\|x_n\| \leq Ca_n$, then with the constant $C' = \max\{\max_{n \leq N} ||x_n||/a_n, C\}$, we get $\|x_n\| \leq C'a_n$ for all $n \in \mathbb{N}$. Some authors do use that definition, but this is equivalent so nothing is lost.

The notation $\mathcal{O}$ is an upper bound, not necessarily the sharpest order. That is, if $x_n = \mathcal{O}(n^{-1})$, then also $x_n = \mathcal{O}(n^{-1/2})$, and even $x_n = \mathcal{O}(1)$.

Proposition 1.26 (Deterministic orders: algebra).
Let $(a_n)_{n\in\mathbb{N}}$ and $(b_n)_{n\in\mathbb{N}}$ be positive sequences. Then, the following relations hold, whenever well-defined.

$$\begin{aligned}
\mathcal{O}(a_n) + \mathcal{O}(b_n) &= \mathcal{O}(a_n + b_n),\\
o(a_n) + o(b_n) &= o(a_n + b_n),\\
\mathcal{O}(a_n)\mathcal{O}(b_n) &= \mathcal{O}(a_n b_n),\\
o(a_n)o(b_n) &= o(a_n b_n).
\end{aligned}$$

Furthermore,

$$\begin{aligned}
x_n = \mathcal{O}(a_n), \text{ and } a_n = o(b_n) \quad &\Longrightarrow \quad x_n = o(b_n),\\
x_n = o(a_n), \text{ and } a_n = \mathcal{O}(b_n) \quad &\Longrightarrow \quad x_n = o(b_n).
\end{aligned}$$

1.3.2 Stochastic extensions of order notations

Definition 1.27 (Boundedness in probability and stochastic order symbols)**.**
A sequence of random vectors $(X_n)_{n\in\mathbb{N}}$ is ***bounded in probability*** or ***tight***, denoted $X_n = \mathcal{O}_{\mathbb{P}}(1)$, if, for all $\epsilon > 0$, there exist $M = M(\epsilon)$ and $N := N(\epsilon)$ such that, for all $n \geq N$,

$$\mathbb{P}\left[\|X_n\| \leq M\right] \geq 1 - \epsilon.$$

Let $(a_n)_{n\in\mathbb{N}}$ be a positive sequence.

- We write $X_n = \mathcal{O}_{\mathbb{P}}(a_n)$ if $X_n/a_n = \mathcal{O}_{\mathbb{P}}(1)$.
- We write $X_n = o_{\mathbb{P}}(a_n)$ if $X_n/a_n \xrightarrow[n\to\infty]{\mathbb{P}} 0$.

Moreover, the other symbols are also extended naturally.

$$\begin{aligned}
X_n \sim_{\mathbb{P}} Y_n \quad &\iff \quad \frac{X_n}{Y_n} \xrightarrow[n\to\infty]{\mathbb{P}} 1, \\
X_n \lesssim_{\mathbb{P}} a_n \quad &\iff \quad X_n = \mathcal{O}_{\mathbb{P}}(a_n), \\
X_n \gtrsim_{\mathbb{P}} a_n \quad &\iff \quad X_n^{-1} = \mathcal{O}_{\mathbb{P}}(a_n^{-1}), \\
X_n \asymp_{\mathbb{P}} a_n \quad &\iff \quad X_n \lesssim_{\mathbb{P}} a_n \text{ and } X_n \gtrsim_{\mathbb{P}} a_n.
\end{aligned}$$

We also write $X_n = o_{\mathbb{P}}(1)$ if X_n converges in probability to zero. When X_n is deterministic, the above definition collapses with the previous one about deterministic sequences.

Proposition 1.28.
The following statements hold.

(i) If $X_n = o_{\mathbb{P}}(a_n)$, then $X_n = \mathcal{O}_{\mathbb{P}}(a_n)$.

(ii) If $X_n/a_n \xrightarrow[n\to\infty]{\mathcal{L}} X$ for some random vector X, then $X_n = \mathcal{O}_{\mathbb{P}}(a_n)$.

(iii) If, for some $q > 0$,
$$\mathbb{E}\left[\|X_n\|^q\right] = \mathcal{O}(a_n^q),$$
then $X_n = \mathcal{O}_{\mathbb{P}}(a_n)$.

(iv) If, for some $q > 0$,
$$\mathbb{E}\left[\|X_n\|^q\right] = o(a_n^q),$$
then $X_n = o_{\mathbb{P}}(a_n)$.

(v) If $X_n = \mathcal{O}_{\mathbb{P}}(a_n)$ with $\lim_{n\to\infty} a_n = 0$, then $X_n = o_{\mathbb{P}}(1)$.

Proof. Proof of (i). Since $X_n/a_n \xrightarrow[n\to\infty]{\mathbb{P}} 0$, for all $\epsilon > 0$, for all $\delta > 0$, there exists N such that for all $n \geq N$,

$$\mathbb{P}\left[\left\|\frac{X_n}{a_n}\right\| > \epsilon\right] < \delta.$$

Hence, the sequence is tight.

Proof of (ii). The Portmanteau theorem can be used to show this; this is left as an exercise.

Proof of (iii). By Markov's inequality,

$$\mathbb{P}\left[\|X_n\|^q > Ma_n^q\right] \leq \frac{\mathbb{E}\left[\|X_n\|^q\right]}{Ma_n^q} \leq \frac{C}{M}.$$

Hence, letting $M = C/\epsilon$, the result follows.

Proof of (iv). Follows again from Markov's inequality.

Proof of (v). Let $\eta > 0$ and let M satisfy

$$\sup_{n\geq 1} \mathbb{P}\left[\|X_n\| > Ma_n\right] < \eta.$$

Since $Ma_n \to 0$ as $n \to \infty$, we have $Ma_n < \epsilon$ eventually, for any $\epsilon > 0$. Hence, we have the inclusion

$$\{\|X_n\| > \epsilon\} \subseteq \{\|X_n\| > Ma_n\},$$

from which we get

$$\mathbb{P}\left[\|X_n\| > \epsilon\right] \leq \mathbb{P}\left[\|X_n\| > Ma_n\right] < \eta$$

eventually, from which the result follows.

■

Similar algebra to that of the deterministic case also applies to stochastic orders.

Proposition 1.29 (Stochastic orders: algebra)**.**
Let $(a_n)_{n\in\mathbb{N}}$ and $(b_n)_{n\in\mathbb{N}}$ be positive sequences. Then, the following relations hold.

$$\begin{aligned}
\mathcal{O}_{\mathbb{P}}(a_n) + \mathcal{O}_{\mathbb{P}}(b_n) &= \mathcal{O}_{\mathbb{P}}(a_n + b_n),\\
o_{\mathbb{P}}(a_n) + o_{\mathbb{P}}(b_n) &= o_{\mathbb{P}}(a_n + b_n),\\
\mathcal{O}_{\mathbb{P}}(a_n)\mathcal{O}_{\mathbb{P}}(b_n) &= \mathcal{O}_{\mathbb{P}}(a_n b_n),\\
o_{\mathbb{P}}(a_n)o_{\mathbb{P}}(b_n) &= o_{\mathbb{P}}(a_n b_n).
\end{aligned}$$

Furthermore,

$$\begin{aligned}
X_n = \mathcal{O}_{\mathbb{P}}(a_n),\ \text{and}\ a_n = \mathcal{O}(b_n) \quad &\Longrightarrow \quad X_n = \mathcal{O}_{\mathbb{P}}(b_n),\\
X_n = \mathcal{O}_{\mathbb{P}}(a_n),\ \text{and}\ a_n = o(b_n) \quad &\Longrightarrow \quad X_n = o_{\mathbb{P}}(b_n),\\
X_n = o_{\mathbb{P}}(a_n),\ \text{and}\ a_n = \mathcal{O}(b_n) \quad &\Longrightarrow \quad X_n = o_{\mathbb{P}}(b_n).
\end{aligned}$$

The symbol $\mathcal{O}_{\mathbb{P}}(1)$ does not mean almost surely bounded; for example, any random variable is $\mathcal{O}_{\mathbb{P}}(1)$ such as Z if $Z \sim \mathcal{N}(0,1)$, yet the support of a Gaussian is $\mathbb{R}$. For a random vector, with fixed dimension, it holds that $X_n = \mathcal{O}_{\mathbb{P}}(a_n)$ if and only if $X_{n,j} = \mathcal{O}_{\mathbb{P}}(a_n)$ for every $j = 1, ..., d$. This holds also for random matrices, entry-wise.

Exercise 1.12 (★★)**.**
Suppose $r_n\left(T_n - \theta\right) \xrightarrow[n\to\infty]{\mathcal{L}} Z$, where Z is non-degenerate[a]. Show that

$$T_n - \theta = \mathcal{O}_{\mathbb{P}}(r_n^{-1}), \quad \text{but} \quad T_n - \theta \neq o_{\mathbb{P}}(r_n^{-1}).$$

[a] A random variable Z is said to be ***non-degenerate*** if Z is not almost surely constant.

The above exercise shows that a non-degenerate normalized limit provides a rate of convergence, which cannot be improved. This is an important notion which we formalize now.

Definition 1.30 (Rate of convergence).
Suppose $X_n \xrightarrow[n\to\infty]{\mathbb{P}} X$. We call a sequence $(a_n)_{n\in\mathbb{N}}$ the ***rate of convergence*** of X_n to X if there exists a non-degenerate random vector Z such that

$$a_n (X_n - X) \xrightarrow[n\to\infty]{\mathcal{L}} Z.$$

With the above definition, the rates are unique up to constant multiplication.

Exercise 1.13 (★★★).
Let $\theta \in \mathbb{R}$ and suppose that $\sqrt{n}\left(\widehat{\theta}_n - \theta\right) \xrightarrow[n\to\infty]{\mathcal{L}} \mathcal{N}(0, \sigma^2)$ with $\sigma^2 > 0$.

(a) Show that $\widehat{\theta}_n^2 - \theta^2 = 2\theta\left(\widehat{\theta}_n - \theta\right) + \mathcal{O}_{\mathbb{P}}(n^{-1})$.

(b) Deduce that, if $\theta \neq 0$, $\sqrt{n}\left(\widehat{\theta}_n^2 - \theta^2\right) \xrightarrow[n\to\infty]{\mathcal{L}} \mathcal{N}(0, 4\theta^2\sigma^2)$.

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